



Research Report

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The Transatlantic Artificial Intelligence Calculus

U.S.–European Union and Member State AI Cooperation
in an Era of Strategic Competition

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About This Report

As global artificial intelligence (AI) competition intensifies, the United States faces critical decisions about how to structure AI partnerships with allies and partners, particularly with the European Union (EU) and EU member states that have developed valuable yet often overlooked AI capabilities. In this report, I provide U.S. policymakers and industry leaders with a strategic framework for assessing the conditional value of AI partnerships across different technological futures and identify options for transatlantic cooperation that advance U.S. strategic interests. This analysis is intended for policymakers and industry stakeholders involved in AI strategy, international technology cooperation, and alliance management.

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Summary

Issue

This study examines scenario-based approaches to structuring U.S.–European Union (EU) and member state artificial intelligence (AI) cooperation amid intensifying U.S.–China rivalry, at a moment when AI leadership is becoming central to economic competitiveness, global standard-setting, and the maintenance of democratic governance (as articulated in *Winning the Race: America’s AI Action Plan* and the 2025 *National Security Strategy of the United States of America*).¹

The case for recalibration does not rest on the absence of transatlantic institutional architecture. Government mechanisms, research, and private-sector partnerships are already substantial, although some have gone dormant. It rests on a scope mismatch between what that architecture delivers and what the competitive moment requires. The existing toolkit enables bilateral dialogue, research collaboration, venture capital for the most visible EU firms, and market access with reasonable competence. What it has not produced is a strategic partnership for AI. For the purposes of this analysis, I define a *strategic AI partnership* as a formal, durable collaboration that aligns resources, rules, and incentives across the AI development and deployment life cycle, enabling each partner to gain capabilities or leverage that it could not efficiently secure on its own while providing a degree of insulation from short-term political or commercial volatility.

The existing architecture has not accounted for the possibility that the strategic value of EU and member state offerings may differ as AI develops. A strategic partnership designed without that contingency thinking risks being well calibrated for one future and poorly positioned for several others, leaving allied capabilities underdeveloped in domains that may prove consequential to the democratic bloc’s collective position. To inform this assessment, I analyze four archetypal scenarios—Concentrated Dominance, Commoditized Frontier, Dependent Specialization, and Collaborative Specialization—representing the structural pathways along which AI development could unfold, defined by two variables: (1) the ultimate locus of AI value creation and (2) the degree of advanced model openness.

By mapping EU and member state capabilities against these potential futures, I propose a conditional framework for partnership, one that moves away from asking how deeply the United States should engage with such allies toward asking under what conditions different forms of engagement generate the greatest returns for U.S. strategic interests. The framework encompasses five cooperation efforts. Two foundational efforts—an evaluation, testing, and standards interoperability stack and lithography resilience efforts—generate returns across all trajectories. The remaining three scenario-contingent efforts are those whose priorities are calibrated against the signposts identified during the course of this analysis.

¹ See Executive Office of the President, 2025; and White House, 2025.

Study Approach

To address the uncertainties inherent in assessing Europe’s role in future AI development, I combined the mapping of current EU and member state AI capabilities, as of June 2026, with an evaluation of how those capabilities might perform across different technological development pathways. Data collection involved collating and cross-referencing materials from multiple source categories: government policy documents, including the EU AI Act; industry reports from technology companies and consulting firms; academic publications identified through targeted searches, using such terms as “transatlantic AI cooperation” and “EU AI governance”; business intelligence from financial news outlets and market research firms; and corporate communications, including press releases. I prioritized primary sources over secondary aggregations.

Stakeholder consultations with private-sector stakeholders, AI researchers, and U.S. and EU current and former government officials provided contextual information that complemented the document-based research. These consultations served two purposes: validating preliminary assessments of EU and member state capabilities against real-world experience and identifying cooperation challenges and opportunities not evident in public sources. The discussions focused on areas in which EU governance expertise, member state sectoral specialization, and institutional deployment capabilities could complement U.S. technological leadership, as well as on structural constraints that might limit the effectiveness of partnerships.

I used AI tools selectively in this analysis to support specific methodological tasks. I used Google NotebookLM as a navigation tool to identify relevant sections within large-scale reports, such as the annual *AI Index Report* published by Stanford University, which I then verified and extracted independently. OpenAI’s Deep Research mode helped identify potentially authoritative sources, all of which I independently verified before including in this analysis. In both cases, AI tools served as research aids rather than as analytical decisionmakers. I retain full responsibility for the interpretation, synthesis, and conclusions drawn from the material underlying the analysis presented in this report.

Building on the empirical foundation from data collection and stakeholder input, I constructed four archetypal scenarios to assess how EU and member state capabilities might perform under different AI development futures. These scenarios derive from two variables that I assessed to be the most uncertain drivers of where AI value may accrue. Finally, I synthesized insights to develop a cooperation framework based on varying assumptions about technological development trajectories, capability evolution, and available resources.

A note on indicator selection is warranted. National and regional AI capacity can be assessed across a variety of indicators, including AI model use, adoption rates, import data, research outputs, and patent activity, among others, and a comprehensive assessment of each was outside the scope of this study. This study examines the EU AI ecosystem across governance frameworks, infrastructure efforts, and model development strengths, which stakeholder consultations consistently identified as the dimensions that most directly bear on the strategic partnership questions that this study examines. Several other indicators appear where relevant throughout the report to paint a broader picture of EU AI capacity, but I did not analyze them in depth as I did the primary inputs to the partnership assessment. Similarly, an in-depth assessment of U.S. and EU export control regimes and AI development for dual-use or military purposes falls outside the scope of this study.

Key Findings

- *According to the scenario logic, two cooperation efforts are likely to generate returns across all four AI trajectories analyzed and should be pursued now.* These are establishing a permanent joint infrastructure for evaluating AI systems and maintaining interoperable safety standards and a coordinated framework to sustain the research and development (R&D), supply chain integrity, and operational security of the allied lithography base.
- *Three additional efforts should be calibrated against observable signposts indicating which of two variables driving AI futures may be materializing: the benchmark gap between proprietary and open-source models and the margin differentials between general-purpose and specialized AI providers.* These efforts involve structuring U.S. private investment to extend compute access to EU startups, small- to medium-sized enterprises (SMEs), and research institutions; deepening cooperation on the privacy-preserving technologies that enable trustworthy AI deployment in regulated sectors; and building financing and distribution networks that help specialized EU firms commercialize. Such a split gives policymakers a defensible sequencing logic: Act on the two foundational efforts now; build institutional readiness for the three scenario-contingent efforts.
- *The EU AI Act's effects run counter to the prevailing narrative.* The often-cited compliance burden is driven by the cumulative effects of the General Data Protection Regulation, sectoral rules, and national policies rather than by the Act itself. To the extent that global AI regulation continues converging toward EU-style risk-classification frameworks, the firms bearing the highest compliance costs are also those best positioned to export compliant solutions, because the conformity infrastructure they have built can be redeployed into new markets at a fraction of the cost that competitors must incur to build one from scratch. The Act should therefore be understood as a conditional first-mover dynamic for trustworthy deployment, with implications for how U.S. firms should think about EU partners as compliance bridges into regulated markets adopting equivalent frameworks.
- *U.S. private capital is reshaping EU member states' AI infrastructure, but its allocation patterns may foreclose broader access outcomes that the alliance should seek to secure.* Hyperscaler-anchored capacity is being absorbed before it reaches the market, often concentrating compute among large enterprise customers and underserving SMEs, startups, and research institutions, where specialized innovation also occurs.
- *Export controls on ASML's sales to China are necessary, but they impose revenue pressure on the R&D pipeline that the U.S. AI hardware base benefits from.* Coordinated long-horizon R&D funding commitments to sustain high numerical aperture extreme ultraviolet lithography innovation across project cycles are largely absent; no public evidence of a pre-agreed U.S. emergency allocation arrangement commits ASML output during a supply crisis; and the publicly available record does not document an ASML-specific allied framework for joint threat intelligence and incident response, although visible coordination has a deterrence and alliance cohesion value that arrangements beyond public view cannot fully substitute for. Together, these vulnerabilities weaken the resilience of one of the most consequential global hardware chokepoints.

- *EU and member state skepticism of U.S. dependency is neither irrational nor hostile; it reflects a legitimate response to accumulated asymmetry.* The same imbalances that provide the United States with leverage today are generating political pressure that may erode it tomorrow, with EU entities increasingly inclined to build alternatives wherever feasible. Maximizing short-term leverage from EU dependence is therefore self-defeating; structuring the relationship so that allied capability grows through cooperation is the most durable insurance against ecosystem fragmentation.
- *EU and member state AI asset value in part depends on whether economic value concentrates in general-purpose or specialized models and on how accessible advanced models become.* For instance, in a future where specialized models capture value comparable to or greater than that of general-purpose systems and open-source models remain competitive enough to serve as accessible foundations for that specialization, EU member states' sectoral expertise becomes one of the principal inputs to AI value creation. By contrast, in a future where a small number of firms control proprietary general-purpose models capable of performing most economically valuable tasks and open-source alternatives fall significantly behind, the EU and member states lose some of their differentiating value.

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Introduction

Global competition over artificial intelligence (AI) supremacy has emerged as a defining contest of the 21st century. The United States commands the high ground in frontier models and hosts the world's leading compute infrastructure (Kariuki, 2025). China is staking its AI leadership claim through state-directed investments, particularly in compute efficiency (Heim, 2025b). The leading nation may shape the rules governing emerging technology, capture disproportionate economic returns, and influence how AI systems reshape the global order (Mazarr, 2026, p. v). Failure to maintain that leadership position risks ceding influence to adversaries (Executive Office of the President, 2025).

The contours of that competition, and the imperatives it generates for U.S. strategy, are by now broadly accepted across the U.S. policy community. Less well-resolved is a question on which U.S. strategy depends: How should the United States partner with its European allies—primarily the European Union (EU) and its member states—to maintain and extend its competitive position?² The question has received substantial policy attention. Whether the existing answers remain adequate given what is now at stake is the subject of this analysis.

Cooperation mechanisms are substantial. The United States, the EU, and member state governments have built scaffolding for technology cooperation across government, university, and private-sector channels, although some of it has gone dormant in recent years.³ Bilateral dialogue, agenda-setting, research collaboration, venture capital for the most visible firms within the EU, and market access all function reasonably well (Schörner, 2026; Amazon Web Services, undated-b). What that architecture has not produced is a result commensurate with what the competitive moment demands.

What would be required is a *strategic AI partnership*: a formal, durable collaboration that aligns resources, rules, and incentives across the AI development and deployment life cycle, enabling each partner to gain capabilities or leverage that it could not efficiently secure on its own while providing insulation from short-term political or commercial volatility. The existing toolkit is considerably weaker at extending scale-up financing to the broader population of specialized EU firms whose domain expertise is sophisticated but whose visibility to U.S. investors is limited (Schörner, 2026; StepUp StartUps Consortium, 2025, pp. 13, 45; European Investment Bank and European Patent

² In this report, I use *Europe* to refer broadly to the EU and its member states, the United Kingdom (UK), and Switzerland. The EU and its member states are the primary unit of analysis given the scale of EU-level regulations and investment frameworks. The UK and Switzerland appear where their experiences are analytically relevant. Otherwise, I have noted where the analysis draws on data or arguments that apply specifically to the EU rather than to Europe more broadly.

³ See, for examples, European Commission, undated-c; European Commission, 2023b; Allen and Benson, 2023; Shivakumar, Wessner, and Howell, 2024; CEPA International Leadership Council, 2025; Szczepański, 2024; U.S. Department of State, 2010; European Commission, undated-f; Imec, undated-a; CIVIS, 2024; College of Europe, undated; Intel, 2012; and Oltermann, 2026.

Office, 2024, p. 2; Draghi, 2024, p. 6). U.S. hyperscaler investment is expanding EU data center capacity at a scale no public program could match, but its current allocation patterns concentrate that capacity among large enterprise customers, underserving the startups, small- to medium-sized enterprises (SMEs), and research institutions where much specialized innovation occurs (Dunn, 2025; Forrester, 2025). The institutional mechanisms for aligning governance approaches—the connective tissue that would enable joint evaluation, shared safety standards, and interoperable risk frameworks—either have fallen dormant or operate episodically, dependent on political will rather than on a durable mandate.

The diagnosis, then, is a mismatch of both scope and durability. Existing mechanisms have not produced, and are structurally unlikely to produce, the level of coordination required to permanently protect irreplaceable allied assets, build shared infrastructure for AI governance, or enable the commercialization of allied capabilities at the scale their strategic potential warrants. Those gaps carry a cost. The collective position the democratic bloc must maintain against state-directed competitors erodes wherever allied assets go unprotected, shared infrastructure frays, or complementary capabilities remain subscale. This analysis proceeds from that diagnosis. Its purpose is to offer a framework for determining, across a range of plausible technological futures, which forms of AI cooperation advance U.S. strategic interests—and, in doing so, strengthen the AI alliance that the United States shares an interest in fortifying (Executive Office of the President, 2025, p. 22).

One aspect of that last point requires emphasis: *a range of plausible technological futures*. This is where the standard policy treatment of transatlantic AI cooperation falls short. The prevailing approach asks how deeply the United States should engage with allies, as though the answer were fixed. It is not. Along certain development trajectories, EU and member state capabilities—particularly in such sectors as health care, manufacturing and robotics, financial services, and government services—could considerably strengthen the U.S. position.⁴ Combined U.S. and European markets dominate digitally enabled service exports and host dense networks of AI talent and research institutions, creating natural advantages for coordinated development (World Trade Organization, 2026; Hook, 2025, p. 1). Under other trajectories, the strategic rationale for deeper engagement weakens considerably. The value of EU and member state AI assets depends on how AI itself evolves—on whether economic value concentrates in general-purpose or specialized models and on whether advanced models remain proprietary or become openly available.

The more defensible question, therefore, is not *how much* to cooperate but *under what conditions* different forms of cooperation generate the greatest strategic returns. Answering that question requires a framework built for uncertainty rather than optimized for a single predicted future. A partnership designed without contingency thinking risks being well calibrated for one trajectory and poorly positioned for several others, leaving allied capabilities underdeveloped in domains that may prove consequential.

I construct that framework in three steps. First, I assess the EU’s current AI ecosystem—governance frameworks, infrastructure conditions, and sectoral applications—to establish what U.S.

⁴ A full comparative analysis of national-level differences in partnership feasibility and strategic risk is outside the scope of this study. Where I make general arguments about European capabilities, constraints, or risks in this report, they should be understood as describing central tendencies across a heterogeneous landscape rather than uniform conditions, and national variation is noted where it is most consequential for the analysis.

allies bring to a partnership and where their structural constraints lie. Second, I develop four archetypal scenarios representing the development pathways along which AI could unfold, defined by two variables whose resolution remains genuinely contested: where AI value ultimately concentrates and how accessible advanced models become. Third, I map EU capabilities against those futures to derive a cooperation framework with two tiers—foundational efforts whose returns are scenario-invariant and contingent efforts whose priority rises or falls with observable signals indicating which trajectory is materializing.

The scenarios share a common baseline assumption that the EU and member states continue along their current AI development path, remaining structurally behind the United States and China in frontier model capacity and deployment. That assumption is a starting point of assessment, not a judgment on European potential. Throughout the analysis, U.S. leverage from current capacity asymmetries should be understood as a window for structuring a durable partnership rather than as an instrument for extracting near-term concessions. That the U.S.–EU and member state relationship is under considerable strain and that this analysis must retain its value regardless of whether such tensions prove durable or temporary are conditions this framework is explicitly designed to accommodate.

The Strategic Inventory

Bold ambitions, coupled with systemic impediments, characterize the EU and member state AI ecosystem. The EU and its member states hold AI assets of genuine strategic value, including a governance architecture with first-mover potential in regulated markets, a monopoly position in advanced lithography, deep sectoral expertise, and a research base of global standing. Those assets coexist with structural constraints—capital scarcity during growth and late-stage development phases, insufficient sovereign compute, fragmented data access, and talent outflow—that have consistently prevented the broader base of AI capabilities from reaching commercial scale. This chapter documents both sides of that picture.

That picture is visible in aggregate metrics. AI deployment is becoming embedded in EU member state economies—20 percent of all EU enterprises used AI in 2025—but the pipeline from research to commercial scale continues to leak (Eurostat, 2026a, pp. 13, 16).⁵ The EU produced 18 percent of global AI scientific publications between 2020 and 2024 yet holds less than 3 percent of global AI patents—a gap the EU, the European Investment Bank (EIB), and the renowned Draghi report have each attributed to fragmented regulation, shallow late-stage financing, and slower technology diffusion (Niestadt, 2025, pp. 3, 10; Schörner, 2026; StepUp StartUps Consortium, 2025, pp. 13, 45; European Investment Bank and European Patent Office, 2024, p. 2; Draghi, 2024, p. 6).

In the sections that follow, I examine the EU’s AI ecosystem across three dimensions: (1) the governance frameworks that shape the operating environment for native and foreign AI firms alike; (2) the infrastructure and resourcing conditions that currently define the scale and accessibility of indigenous AI development, as well as the investments that are beginning to alter that picture; and (3) illustrative sectoral applications in which EU member states have built meaningful, strategically relevant capabilities that complement U.S. strengths. Together, these three dimensions provide the empirical foundation for the scenarios that follow, and their interactions, which I examine at the close of the technology stack assessment, determine which cooperation efforts generate compounding returns.

A Conditional First-Mover

The EU AI Act’s effects run counter to the prevailing narrative. The most documented compliance burden is the cumulative effect of the General Data Protection Regulation (GDPR), sectoral regulations, and national implementation strategies—not solely the EU AI Act (Fairless and

⁵ These figures vary by member state, from 42.0 percent enterprise adoption in Denmark to 5.2 percent in the lowest-performing jurisdictions, but they collectively signal that AI is becoming embedded in member states’ economic life (Eurostat, 2026a, p. 16).

Luhn timer, 2025; Almada, 2025). The Act itself—a risk-classification framework that imposes graduated obligations based on potential harm—subjects only 5–15 percent of AI systems to its heaviest compliance requirements (Laurer, Renda, and Yeung, 2021).

Yet those requirements generate conformity infrastructure—audit workflows, documentation architecture, tested risk-management systems, and trained personnel—whose marginal cost of redeployment into new markets adopting comparable frameworks would be a fraction of the initial development cost. To the extent that global regulation continues converging toward EU-style risk-classification models, the Act creates a conditional first-mover dynamic for trustworthy deployment, with direct implications for how U.S. firms should assess EU partners as compliance bridges into regulated markets.

That condition is partially met and advancing. Brazil’s AI bill, Colorado’s AI legislation, and emerging elements in Asian jurisdictions each show convergence toward EU-style provisions, although none yet replicates the full framework (Almada, 2025). The EU has moved to accelerate the timeline on its own through the AI Pact, which invites companies to implement the Act requirements ahead of legal deadlines, compressing the interval between regulatory adoption and conformity infrastructure maturity (EU, 2026). The trajectory is therefore plausible and partially underway but not yet certain enough to be treated as a planning assumption. It is a conditional asset whose value scales with the number of jurisdictions that adopt equivalent provisions and whose absence would substantially narrow the Act’s strategic significance to the EU’s internal market alone.

The Act’s limited reach is not randomly distributed across the economy. Many AI applications in industrial manufacturing and robotics, a sector where EU member states hold documented comparative advantages, are subject to limited regulation (Laurer, Renda, and Yeung, 2021). For EU member state companies adopting AI, particularly SMEs that use off-the-shelf models in non-high-risk contexts, the Act’s direct obligations remain limited compared with existing GDPR and sector-specific compliance work. Where high-risk classification does apply—notably in health care, financial services, and government operations—it is those very obligations that generate the conformity infrastructure whose first-mover value is described above (European Parliament, 2025; Vardas, Marketou, and Vardas, 2025; European Insurance and Occupational Pensions Authority, 2024). In these sectors, the compliance cost and the strategic asset are the same investment viewed from different angles.

The compliance friction most reported by EU firms reflects the cumulative interaction of multiple regulatory layers rather than the EU AI Act in isolation. Businesses allocate 40 percent of their information technology (IT) budgets to regulatory compliance, and two-thirds remain unclear about the Act’s specific obligations, a confusion that itself suggests the difficulty of disentangling one instrument from the broader stack (Fairless and Luhn timer, 2025). The EIB finds that 59 percent of EU firms view compliance with new regulations as an obstacle to business activities, with bureaucratic costs consuming an estimated 1.1 percent of turnover for firms broadly and 1.8 percent for smaller companies (EIB, 2025, pp. 3, 12). Divergent national implementation strategies compound the picture; despite unified EU legislation, varying member state approaches generate additional friction for firms operating across borders (EU Artificial Intelligence Act, 2026). The cumulative weight of these layers—not any single instrument—is the primary friction point.

That cumulative burden has been associated with observable consequences for market access. Meta and Apple each postponed EU product launches, citing EU regulatory requirements as factors in their decisions (Milmo, 2024; Goswami, 2024; Draghi, 2024; Chee, 2024; Volpicelli, Gurman, and Lepido, 2026). Product-readiness considerations and broader commercial strategy likely also shaped timing, but the companies' stated rationales nonetheless point to friction that constrains U.S. firms' ability to deploy AI capabilities in EU markets. The regulatory environment has also contributed to EU firms' relocation decisions, although it operates as one among several interacting factors, including compensation differentials, capital availability, and market scale (EIB, 2026, p. vii; Latona and Pinedo, 2026; "European Startups Move to the U.S. for Looser Tech Regulations," 2025; Nowshin, 2023).⁶

The EU has built direct countermeasures to the compliance uncertainty documented above. For instance, regulatory *sandboxes*, mandated across all member states, allow firms to test AI systems in controlled environments with direct regulatory guidance before bearing full conformity costs (EU Artificial Intelligence Act, 2025). For SMEs and startups, in particular, sandboxes and other such simplification mechanisms reduce the gap between regulatory intent and operational clarity that the "two-thirds unclear" finding reflects, offering a structured pathway through the overlapping demands of the EU AI Act, GDPR, and sectoral rules rather than requiring firms to navigate them unaided (European Commission, undated-g).

Because the EU AI Act's enforcement timeline is only beginning, definitively assessing its operational impact would be premature. What can be said is that the conformity infrastructure the Act mandates also generates auditable deployment records that function as a standing security screening layer for AI systems entering critical allied sectors, a benefit the United States captures without building or funding these mechanisms independently. For the partnership framework discussed later in this analysis, the EU's governance infrastructure constitutes a conditional asset that cooperation should be designed to leverage, and the cumulative compliance friction constitutes a design constraint that cooperation mechanisms should help manage.

That friction is unlikely to resolve quickly. The European Commission's proposed Digital Omnibus simplification package targets smoother implementation of the EU AI Act, and such figures as French President Emmanuel Macron and former Italian Prime Minister Mario Draghi explicitly advocate regulatory efficiency (Scott, 2025; Merli, 2025). Yet securing agreement among the 27 member states remains challenging; reform tends to succeed only when national priorities align with EU objectives, leaving most simplification efforts stalled in lengthy negotiation (Schumann, Minano, and Maggiore, 2025; European Council, 2026).

⁶ Comparable dynamics have been observed among firms operating under the UK's post-Brexit regulatory landscape. The UK's 11x, for instance, relocated to the United States, citing regulatory pressures, reflecting that the structural conditions discouraging firm retention are not confined to EU member states (Nicol-Schwarz, 2024). The UK and Switzerland, while not subject to the EU AI Act, face their own governance pressures. The UK is developing an AI regulatory framework as part of its post-Brexit legislative agenda, while Switzerland maintains its own data protection regime under the revised Federal Act on Data Protection. Where these frameworks intersect with EU requirements, particularly for firms operating across multiple European jurisdictions, adds further layers of compliance complexity relevant to U.S. firms seeking to operate across the broader European market. However, a systematic treatment of UK and Swiss regulatory dynamics is outside the scope of this discussion.

The Infrastructure Beneath: From Lithography to Talent Retention

To identify where deeper engagement can most effectively support capability development, we must first understand the EU and member states' position across relevant components of the *technological stack*—the layered infrastructure of hardware, software, and services that enables AI systems to function. Central to this effort is the EU's EuroStack initiative, an ambitious digital sovereignty project aiming to reduce dependence on external technology providers by building indigenous infrastructure across semiconductors, cloud services, AI platforms, and data governance frameworks (EuroStack, undated). However, EuroStack faces formidable obstacles, including a projected cost exceeding €5 trillion for a fully independent stack, market fragmentation across EU member states, and balancing sovereignty goals with geopolitical and market dynamics across components (Cheslow and Pollet, 2025).

Rather than surveying that full ambition, the subsections that follow provide an overview of the dimensions of the stack that most directly bear on cooperation design: ASML's lithography machines, EU compute and data access conditions, the capital environment that finances development across all layers, and talent.⁷

ASML's Next-Generation Lithography Monopoly

The most unambiguous global leverage within the EU resides in advanced semiconductor manufacturing equipment, where the Dutch-based ASML holds a monopoly on high numerical aperture extreme ultraviolet (High-NA EUV) lithography—the only technology capable of producing the most-advanced chips powering frontier AI systems (ASML, undated-a; van der Lugt, 2024). The magnitude of this position warrants emphasis. ASML is expanding its annual output by 36 percent in 2026, a ramp-up whose pace is set not by demand but by the production capacity the company can physically bring online (Mackrael, 2026).

ASML's research and development (R&D) spending is also growing by roughly 16 percent annually and will likely need to accelerate further (Mugisha and Jenkins, 2025, p. 3). That acceleration is largely driven by AI customers who are seeking a 16-fold increase in the density of transistors every two years, up from a historical twofold increase over the same time frame, a trajectory whose achievability, according to ASML's chief executive officer, depends on the company's ability to fund the extraordinarily high cost such a pace demands (Norges Bank Investment Management, 2025, minutes 7–9). Because ASML's R&D is primarily financed through commercial revenue, sustained high demand can partially close that gap (ASML, 2025a, pp. 33, 52). Nonetheless, the gap between what AI customers require and what revenue alone can support is not trivial. ASML has, in the past, asked customers to co-invest directly in its R&D to sustain the pace of innovation (Norges Bank Investment Management, 2025, minute 13; ASML, 2012a).

ASML's R&D model functions well when its customer base is broad and expanding. It comes under pressure when that base contracts. Since 2018, successive rounds of U.S. export restrictions

⁷ This analysis treats capital as a layer of the technology stack rather than as an external economic condition. That framing reflects an emerging consensus that *investment infrastructure*—the availability, scale, and terms of financing—functions as a binding constraint on AI development in the same manner as compute or data access and should be analyzed accordingly (Girishankar et al., 2025).

have narrowed ASML’s addressable market by limiting sales to China, reflecting lithography’s central role in broader technology competition (van der Lugt, 2024). The gradual expansion of these controls, potentially extending to ASML’s standard deep ultraviolet systems under proposed legislation, such as the Multilateral Alignment of Technology Controls on Hardware Act, constitutes a staggering 48 percent of the company’s revenue (Bermingham, 2026; Roy, 2026). Each incremental restriction carries a security rationale; each also reduces, absent offsetting mechanisms, the revenue available to sustain the innovation pipeline on which allied technological advantage rests. The tension is not between security and commerce in the abstract but between two allied objectives—constraining adversary access and preserving the capacity that makes allied technology worth protecting—that current policy architectures do not yet reconcile.

Some efforts to support ASML’s growth exist. Among these, the €2.5 billion Project Beethoven, backed by the Dutch government, facilitates the broader industrial expansion required to meet demand (Houtzager, 2024). It is not, however, a direct R&D subsidy; within the EU, direct state aid of that kind is heavily regulated and requires extensive deliberation for approval (Houtzager, 2024). Rather, Project Beethoven invests in output capacity. More capacity sustains higher supply, which generates more revenue, and more revenue finances forward research. What remains absent is a transatlantic mechanism to ensure that allied security architectures actively account for the innovation costs they impose on the firms they exist to protect.

Narrowing the Compute Gap

Despite sizable EU-level commitments, including the €43 billion European Chips Act and the €200 billion InvestAI initiative, Europe accounts for only 4–5 percent of global compute capacity, and the volume of available graphic processing units (GPUs) falls short of demand (European Commission, 2026d; European Commission, 2025a; Doan, Levy, and Storch, 2025, p. 3).⁸ At the funding level, a small proportion of these figures reflects directly committed EU budget allocations. The remainder relies on EU member state contributions and private-sector investment, both unpredictable, given competing national priorities and domestic investors’ aversion to the long-horizon returns that AI infrastructure projects offer—although foreign capital commitments have helped compensate for the shortfall.

At the hardware level, the European Chips Act’s focus on mature rather than leading-edge semiconductor nodes was a deliberate strategic choice, made in consultation with U.S. policymakers and companies, on the grounds that older nodes, such as the 28/22 and 16/22 nanometer levels at which Taiwan Semiconductor Manufacturing Company’s (TSMC’s) €10 billion Dresden facility will operate, help secure Europe’s semiconductor supply chain resilience and fulfill European customers’ production needs (TSMC, 2024; European Commission, 2024b). Intel’s €80 billion Magdeburg commitment similarly targets supply chain segments where European industrial integration generates sustainable demand (Intel, 2023). Where EU member state firms can access leading-edge compute through trade while concentrating domestic capacity in sectors of comparative advantage, that division

⁸ As AI systems become more agentic, their compute footprint has been evolving, with central processing units reassuming a meaningful role in orchestration and inference-time tasks, even as GPUs remain a critical node for model training and deployment at scale.

of labor follows a clean logic of trade specialization—but it also means that domestic fab investment, however large in headline figures, does not close the leading-edge compute gap that frontier AI development requires.

That picture is beginning to shift materially. Multiple European High-Performance Computing Joint Undertaking (EuroHPC JU) AI gigafactories, including Jupiter in Germany and IT4LIA in Italy, have come online and explicitly prioritize SMEs and startup access, helping address the gaps left unresolved by earlier EuroHPC JU projects (EuroHPC JU, 2025; EuroHPC JU, 2026). Alongside these public-private efforts, U.S. private capital has been actively shaping EU AI infrastructure. Microsoft’s planned \$10 billion data center in Portugal, Google’s €5.5 billion investment in data center capacity in Germany, and Amazon’s almost €34 billion Spanish cloud capacity expansion represent only a fraction of the numerous U.S. investments in EU member state data centers to date (Dunn, 2025; Google Cloud, 2025; Latona and Pinedo, 2026). Together, these developments will broaden compute capacity, although many will also exacerbate EU reliance on hyperscaler tenants, who control more than 70 percent of the European cloud market (Gineikyte-Kanclere, Eggert, and Skiotyte, 2025, p. 1).⁹

The hyperscaler investment reshaping EU infrastructure is also unevenly distributed across the continent and primarily benefits large enterprise customers (Forrester, 2025). Ireland, Portugal, Spain, and Nordic countries, among others, continue to develop formidable capacity relative to their size, in part because of their energy advantages, a pattern that underscores how heavily the case for building data centers turns on national context (Sachs and Saraiva, 2026; Webber, 2026).

The EU’s aggregate sovereign compute position, therefore, remains insufficient for competitive large-scale AI development without foreign anchor tenant investment, and that investment arrives on terms that deepen dependency. The compute constraint constitutes several interacting bottlenecks, among them capital scarcity, energy costs and grid limitations, and hyperscaler leasing patterns that concentrate capacity before it reaches the broader market. Of these, capital is among the most tractable, and it is arriving, albeit unevenly, across the continent. The resulting allocation patterns concentrate capacity among large enterprise customers while underserving startups, SMEs, and research institutions where much specialized innovation occurs, an allocation problem that I address directly in Chapter 5.

Abundant Data, Restricted Flow

Even if the EU and its member states solved their computing challenges, accessible, high-quality data would remain a binding constraint. The raw material is plentiful. EU member states generate strategically distinctive datasets at scale, including longitudinal clinical records within national health systems and granular data on securities, loans, and entities across the EU’s banking sector, that are hard to replicate elsewhere (European Commission, undated-d; Mayerlen, 2023). The problem is not volume or quality; it is accessibility.

⁹ Overall, the compute dependency may be concerning, but the security calculus would look materially different if Chinese rather than U.S. providers held that share; the resulting dependencies could hold considerably more-adverse implications.

Data sources remain fragmented across sectors and borders, with legal and regulatory barriers restricting access to training datasets (Righini and Blythe, 2025). That siloing is compounded by the isolation of the innovation ecosystems that would need to share data in the first place. Divergent national strategies have created regional AI clusters, including in Île-de-France, Oberbayern (Upper Bavaria, Germany), and Noord-Brabant (North Brabant, Netherlands), that contain world-class universities, national laboratories, and private AI firms but operate with limited cross-border integration (Bianchini and Ancona, 2023; Brainport Development, 2021; FIWARE Foundation, 2025). The result is that data assets generated within one hub rarely flow to developers in another, undermining the EU's pan-European cohesion objectives and constraining efforts to scale domestic champions across the continent.

EU policymakers have launched several initiatives to address these challenges. The EU Data Governance Act and EU Data Act seek to clarify data-sharing rules and facilitate the reuse of public and protected data, and the European Commission's Data Union Strategy, Common European Data Spaces, and data labs within AI factories target geographic and sectoral fragmentation by aggregating and sharing datasets across borders (European Commission, 2025b; Big Data Value Association, 2025; European Commission, 2025f; European Commission, 2025i; European Commission, 2024c). Yet legal uncertainty continues to cloud copyright, privacy, and the lawful use of AI training data under GDPR's strict protections.

Unlike the compute constraint, the EU data problem is governance-driven at its core. The most distinctive data assets in the ecosystem fall under GDPR and sectoral rules. None of these barriers yields to capital deployment or compute expansion. Resolution requires regulatory simplification, harmonized national frameworks, and the maturation of privacy-preserving technologies that would allow sensitive data to be used in AI development without legal exposure that currently deters such use.

The Capital Gap: Why EU AI Firms Struggle to Scale at Home

AI investment in Europe has grown in absolute terms, but it is a rounding error against U.S. levels, which are projected to reach roughly \$700 billion by the end of 2026 (Weise, 2026). Global corporate AI investment more than doubled in 2025, reaching nearly \$345 billion, driven primarily by generative AI, although European firms secured just \$20.92 billion compared with \$285.88 billion in the United States (Sajadieh et al., 2026, pp. 173, 179, 183). Between 2024 and 2025, U.S. private AI investment rose by more than \$150 billion, while European investment increased by less than \$5 billion, and Germany, France, Italy, and Spain together saw 182 AI companies funded, compared with 1,953 AI companies in the United States (Sajadieh et al., 2026). These figures include the UK and other non-EU member states, meaning that investment at the EU member state level was more modest still.

The concentration of global investment in generative AI partly explains the gap; frontier AI model development is substantially more capital-intensive than the niche application development in which many European firms specialize. However, the EU's comparatively modest increase, alongside a thin pipeline of funded companies, points to a structural feature of the EU capital landscape. EU investors supply most of the funding in early rounds of under €10 million, but their participation drops to

26 percent in AI deals above €25 million, with most late-stage capital flowing from U.S. and UK sources (European Commission, 2025e). Even with U.S. and UK investors providing the bulk of late-stage financing, the aggregate result still leaves the EU well behind the scale its AI ambitions require, and such financing tends to concentrate among the most visible firms, leaving a long tail of specialized EU firms underserved (Schörner, 2026). Notably, China’s private AI investment, while starting from a lower base, has grown at a faster rate than Europe’s over the same period, pointing toward a convergence that sits uncomfortably alongside U.S. and EU aspirations to anchor democratic AI globally.

The conditions shaping foreign and domestic investment patterns vary across the EU landscape, but several factors may explain why capital remains relatively scarce for late-stage firms, according to stakeholder discussions. Table 2.1 briefly describes each factor.

Table 2.1. Drivers of the Capital Gap

Factor	Open-Source Evidence and Implications
Energy costs and grid access	The cost of electricity in most EU member states runs well above U.S. prices (Energy Information Administration, 2026). Capacity constraints, overreservation, permitting delays, and the “no right to refuse” rule further strain grid limitations (Eurelectric, 2026). Frankfurt’s electricity demand, for instance, already exceeds available capacity (Jungblut, 2025). Capital investors have responded selectively, concentrating on locations where conditions are favorable, leaving several tech hubs in Europe underserved (Popescu and Riley, 2026).
Hyperscaler dependence	AI training and inference functions rely primarily on foreign-owned or leased cloud infrastructure (O’Reilly-Durand, 2025). This dependence weakens the investment case for firms built on top of it, particularly for domestic investors who value sovereign AI (Hodgson, 2026). Newly built data centers will also be absorbed by foreign hyperscaler tenants: Start Campus’s SIN01 facility, though technically EU sovereign cloud, was fully leased to Microsoft before operations began, deepening rather than offsetting reliance on U.S. hyperscalers (Dunn, 2025).
Talent retention	Net tech talent inflow to Europe fell by half between 2022 and 2024, and only 10 percent of European top AI researchers remain on the continent (Kiorri, 2026; Mendel, 2025). The resulting compression of the domestic talent pipeline constrains investment attractiveness (Partington, 2025).
Domestic investment culture	Domestic EU venture capital investors have historically avoided long-horizon, high-risk, high-return investment (Collet et al., 2026). Well-positioned firms therefore depend on U.S. and UK capital to scale, yet foreign inflows have not closed a deep-tech funding shortfall estimated at up to \$24 billion (King et al., 2026). France’s Mistral AI, which relied on U.S. investors to achieve unicorn status, is a quintessential example (Tracxn, 2025). Redistributing just 1 percent of Europe’s pension assets could materially reshape the sector (Olson, 2025).
Perception of burdensome regulation	The Draghi (2024) report, published by the EU, states that “innovative companies [wanting] to scale up in Europe are hindered at every stage by inconsistent and restrictive regulations” (p. 6). The cumulative weight of the EU AI Act, GDPR, and sectoral obligations may narrow the number of ventures perceived as capable of clearing compliance hurdles, further weakening the investment case.

SOURCES: The factors described in this table are drawn from stakeholder consultations conducted for this study, supplemented by cited open-source research. They represent the conditions most identified across those discussions as constraining Europe’s AI investment landscape and are not intended as an exhaustive account of the barriers to capital formation.

To address U.S. hyperscaler dependence, EU member state firms have begun developing indigenous alternatives to foreign cloud infrastructure in response to these constraints. Mistral AI's Mistral Compute, which offers an integrated stack of GPUs, orchestration, and application programming interfaces and serves as a data-sovereign alternative, is the most prominent example (Mistral AI, 2025b). However, its ultimate impact hinges on achieving the scale and pricing necessary to compete with hyperscale cloud providers, a threshold that EU member state infrastructure ventures have historically struggled to reach absent coordinated demand aggregation, anchor procurement commitments, and the kind of sustained domestic capital commitment that has proven most elusive in the European context.

Public development banks fill some of the financing gaps described above. Bpifrance has committed €10 billion through 2029 across the AI value chain, KfW Capital deployed €748 million to fund startups and young technology firms in 2025, and the EIB's InvestAI Initiative seeks to mobilize €200 billion in public-private partnerships, helping anchor co-investment that may not have otherwise occurred (Dillet, 2025; KfW Capital, 2026; European Commission, 2025a). Despite these public efforts, Europe's deep-tech startup funding shortfall is estimated at up to €24 billion (King et al., 2026, p. 3).

French President Macron's strategy of attracting foreign investment has generated headline-scale commitments: The February 2025 Paris AI Action Summit produced €109 billion in announced investment, including Brookfield's €20 billion commitment and the United Arab Emirates-backed MetaGPT X partnership for a €30–€50 billion data center campus near Paris (Brookfield, 2025; Piquard, 2025a). More recently, at the June 2026 Choose France summit, SoftBank committed €75 billion to AI data centers in northern France, and the event attracted €93 billion in total AI investment (Nussbaum and Adghirni, 2026). The causal pathways from those commitments to durable domestic AI capacity, however, rest on conditions that are not uniformly present across EU markets. France's model benefits from abundant nuclear power, a unified regulatory regime, and a national development bank that seeks to anchor co-investment at scale. These are advantages that do not readily translate to EU member states with higher energy costs, shallower institutional infrastructure, or varying national appetites for foreign investment.

A Talent Problem That Is Not a Talent Shortage

Table 2.1 includes talent retention as a factor constraining investment attractiveness; a full account of EU member state talent dynamics lies beyond the scope of this analysis, but one distinction matters directly for cooperation design. One mapping of global AI professionals found that Europe has roughly 30 percent more AI talent per capita than the United States and nearly three times as much as China (Kiorri, 2026). The problem, then, is not production but talent attraction and retention. Net tech talent inflow to Europe fell from around 52,000 individuals to just 26,000 individuals between 2022 and 2024; only 10 percent of the top native AI researchers remain on the continent, and the outflow is disproportionately concentrated among senior-level professionals (Kiorri, 2026; Mendel, 2025).

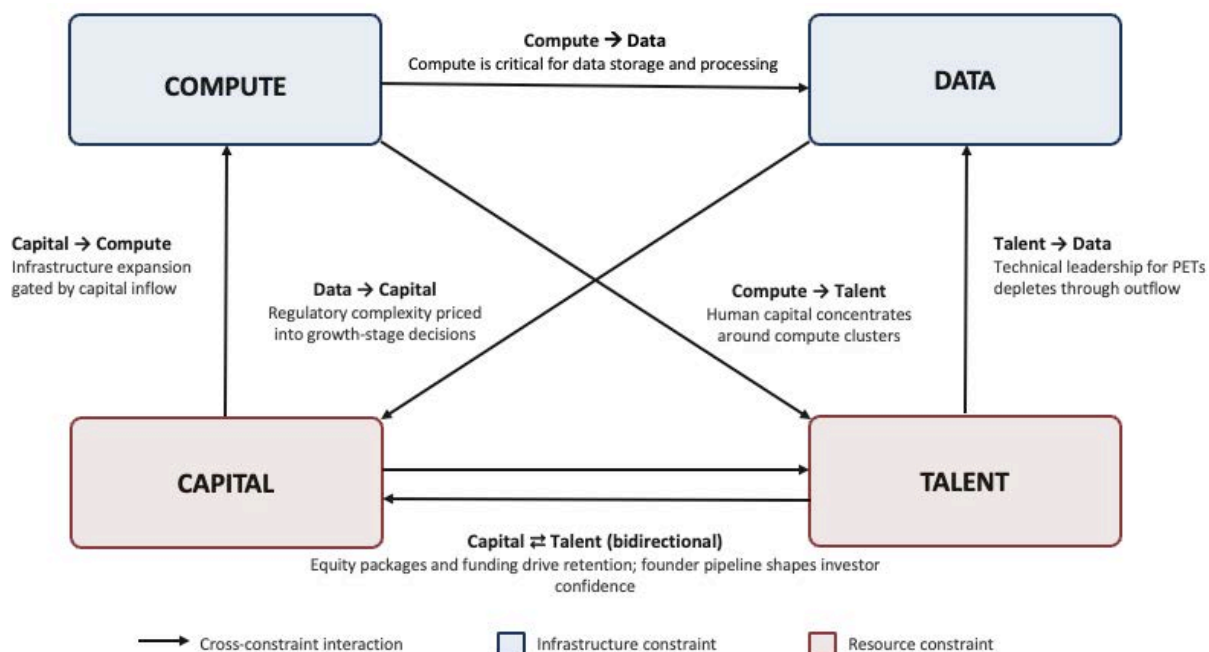
The retention problem operates through multiple channels, whose relative weights are difficult to isolate. Firms may relocate to access deeper venture capital ecosystems and lighter regulatory

environments. Meanwhile, individual talent moves toward U.S. salaries that typically exceed European equivalents by 30–70 percent and equity structures that remain comparatively rare in EU markets (Kiorri, 2026). Stricter labor laws may compound both streams (Organisation for Economic Co-operation and Development [OECD], undated-a). The diversity of drivers matters. Talent outflow is simultaneously a symptom of the conditions described throughout this chapter and a force that deepens them, compressing the domestic pipeline that investors and firms need to scale.

How These Constraints Compound—and What That Means for Cooperation Design

The preceding subsections examined compute, data, capital, and talent individually. Their strategic significance, however, also lies in how they interact, because cooperation efforts that address interactions can generate compounding returns. Figure 2.1 maps the principal interactions among these four factors. The figure includes only constraints that interact bidirectionally and function as direct resource inputs whose scarcity bounds AI development. EU governance frameworks are omitted because they operate as conditioning rules rather than as resources, shaping how capital flows, how data are accessed, and how talent perceives the ecosystem, rather than constraining AI development through their scarcity. ASML’s lithography position is omitted because it constitutes an allied asset to be protected rather than a constraint to be resolved. Both components enter the cooperation framework through a different logic, developed in the following chapter.

Figure 2.1. EU AI Constraint Interactions



NOTE: The figure depicts the interactions most central to this analysis of cooperation design and is not intended as a comprehensive map of all factors shaping EU AI development. Energy costs, regulatory fragmentation across EU member states, public R&D allocation, and other conditions discussed elsewhere in this chapter shape each constraint but are omitted here for the purposes of this analysis. PETs = privacy-enhancing technologies.

These interactions carry consequences for the design of cooperation. First, capital scarcity reinforces talent outflow, as equity packages intended to retain technical personnel depend on growth and late-stage capital, which EU markets supply less readily than U.S. markets. Second, regulatory uncertainty around data may exacerbate capital scarcity because investors factor in compliance complexity when making funding decisions. Third, talent outflow reinforces the data constraint by narrowing the pool of technical personnel needed to build the privacy-preserving deployment infrastructure required to lawfully use sensitive data assets within the EU. Fourth, compute access shapes talent retention because researchers and senior AI professionals gravitate toward environments where they can perform substantive experiments at scale. Fifth, and most consequentially, compute capacity depends on capital investment, energy availability, and grid access, and sufficient compute access is a prerequisite for processing and training on the large-scale datasets that EU data initiatives aim to unlock.

The cooperation efforts developed in Chapter 6 are chosen in part based on whether they address individual constraints or operate on their interactions. Those that touch this interaction loop more directly are likelier to generate compounding returns than those trained on a single constraint alone.

Distributed Excellence in AI Applications

Alongside the metrics of model scale and compute intensity on which U.S. and Chinese competition is most visibly fought, other indicators, including specialized model development, domain-specific research output, and sectoral deployment, tell a meaningfully different and more favorable story about the EU's AI trajectory. The pathway to an economically and strategically consequential AI future does not hinge solely on indigenous frontier-model leadership; it requires the capacity to develop, deploy, and scale AI systems that generate value in the domains where a given economy has the deepest expertise and strongest institutional integration. It is along that pathway that EU member states have built their most durable and strategically significant AI capabilities.

While firms in the EU market cannot match the sheer scale of resources driving U.S. frontier model development, they are quietly generating pools of excellence in select AI applications. This is not a second-best position relative to frontier model development; it is a potentially decisive competitive posture, particularly under technological trajectories where domain-specific models prove as valuable as general-purpose models. France's €10 billion commitment exemplifies this, combining investments in foundation models with targeted funding for AI in health care and industrial automation (Bpifrance, 2025; AI Action Summit, 2025). Germany similarly allocates portions of its €5 billion strategy to sector-specific applications, particularly industrial domains, where German firms already lead technologically (European Commission, 2021).

EU member states possess meaningful competitive advantages in at least four sectors—manufacturing and robotics, health care, financial services, and government operations—although the nature of those advantages varies across them. In manufacturing, the advantage is most clearly documented relative to other geographies. European manufacturers already lead, with 48 percent using AI technologies, compared with 28 percent of U.S. manufacturers as of 2026 (“Europe Can Still Win the Other AI Race,” 2026). In health care, financial services, and government operations, the advantage is less a matter of scale than of depth. EU member state firms have accumulated domain

expertise, regulatory sophistication, and institutional integration in sectors where AI deployment demands those qualities, positioning them as natural development environments for applications that require high trust, auditability, and compliance.

Europe's strong industrial base, combined with deliberate national strategies to position AI as a driver of economic competitiveness, makes these sectors particularly promising environments for AI adoption and application development. Manufacturing exports, with Germany as the largest contributor, are generating trade surpluses and creating deployment opportunities for AI applications across industrial supply chains (European Commission, 2021; Eurostat, 2026b). Similarly, health care applications show substantial promise, with Western European digital health spending projected to reach a €222 billion market by 2030 (Medić, 2025). The health care, government operations, and financial services sectors have particularly strong export prospects to countries that have adopted GDPR-like frameworks or that prefer solutions developed within robust risk-management frameworks.

The capacity for world-changing specialized AI development within the broader European ecosystem is not hypothetical, although the most prominent example to date falls outside the EU's borders. DeepMind, founded in London and now part of Google, produced AlphaFold, which solved the protein-folding problem, transforming structural biology globally (Google DeepMind, undated). Notably, the AlphaFold Protein Structure Database was launched and is maintained in partnership with the European Molecular Biology Laboratory's European Bioinformatics Institute, an arrangement that reflects the kind of institutional infrastructure in which pan-European, interorganizational entities hold comparative strength (Google DeepMind and European Molecular Biology Laboratory-European Bioinformatics Institute, undated).

Within the EU, Bosch's Center for Artificial Intelligence exemplifies a parallel model of systematic, specialized development. Bosch trains its models for factory optimization, anomaly detection, automotive decisionmaking, and supply chain forecasting using proprietary sensor data from its manufacturing operations, producing systems whose strategic value derives entirely from domain expertise and institutional integration (Bosch Rexroth, undated). These examples are not outliers. They represent an AI development model that is increasingly prevalent in EU member states that may prove as valuable as frontier model capacity.

Table 2.2 presents several examples of EU member state firms that are already capitalizing on such sectoral advantages by combining domain expertise with model development, fine-tuning, and subsequent adoption.

Table 2.2. Illustrative EU Member State AI Companies

Sector	EU Member State	Company	Product
Robotics and manufacturing	Germany	Siemens	Digital twins; industrial automation
	Germany	KUKA	Collaborative robots (LBR iisy)
	Denmark	Universal Robots	Collaborative robots (AI Accelerator)
	Italy	Comau	Industrial automation for automotive industry
	Germany	Bosch Rexroth	AI-enabled motion control and smart factory solutions
	Germany	NEURA Robotics	Cognitive robotics
Health care	Germany	Siemens Healthineers	AI-driven medical imaging
	France	Owkin	Health care AI for pharmaceutical R&D
	Germany	ZEISS	Clinical and research data analysis
	Denmark	Corti	Emergency response and patient flow
Government and public services	Germany	Aleph Alpha	Public administration AI systems
	Belgium	ML6	Strategic planning and governance frameworks for AI adoption
	Spain	Flexidao	Energy management
Financial services	France	Mistral AI	Fraud detection, operational efficiency
	Germany	Aleph Alpha	Compliance, operational efficiency
	Spain	Multiverse Computing	Optimization algorithms
	Germany	SAP	Enterprise resource planning

SOURCES: Features information from ABB, undated-a; ABB, undated-b; Aleph Alpha, undated-a; Aleph Alpha, undated-b; Bosch Rexroth, undated; Comau, undated; Corti, undated; Flexidao, undated; Herzig, 2025; Hussain, Kelly, and Surender, 2024; Mistral AI, 2025a; Mistral AI, undated-a; ML6, undated; NEURA Robotics, undated; Owkin, undated; Multiverse Computing, undated; Siemens, undated; Siemens Healthineers, undated; Stäubli, undated; “Top 10 Industrial Robots Companies in UK, Europe, Germany, China, India, US and Japan in 2025,” 2025; Universal Robots, undated; Willmer, 2022; ZEISS, undated.

NOTE: Companies are drawn from EU member states, selected based on (1) market leadership or significant venture backing within their respective sectors and (2) documented AI-specific product offerings or capabilities. The table is illustrative rather than exhaustive. Its purpose is to demonstrate the existence, sectoral breadth, and geographic distribution of specialized AI capabilities within the EU. Selection reflects my assessment of which firms most clearly exemplify the development model described in this section; other firms meeting the stated criteria exist in each sector but were omitted for brevity.

Many companies in Table 2.2 employ a transatlantic division of labor that maximizes the comparative advantages of both blocs. Of the 16 companies examined, 15 maintain active U.S. partnerships spanning investment, distribution, cloud infrastructure, and research collaboration, while EU member state firms contribute specialized R&D and serve as testing grounds. For example, European unicorn Mistral AI's models depend on U.S. tech giants for infrastructure and distribution (Heim, 2025; Mistral AI, undated-b). Non-EU industrial leaders, such as ABB and Stäubli, also leverage U.S. partnerships to expand globally, contributing tens of billions of dollars to the U.S. economy (ABB, undated-c; Stäubli, undated). Aleph Alpha represents a notable exception, embodying the AI sovereignty concerns that are gaining popularity in Europe (Bergen, 2024). Such a model allows European and U.S. firms to enhance returns and provides a foundation for deeper collaboration.

The strategic question for U.S. stakeholders is therefore not whether EU member states possess AI capabilities worth engaging. It is whether the technological trajectory that ultimately prevails is one under which those capabilities generate the greatest returns—a question the four scenarios that follow are designed to answer. Most firms in Table 2.2 have already secured the partnerships required to reach global markets; the broader population of specialized EU firms with comparable domain expertise but less visibility and capital have not—a gap the cooperation framework in Chapter 6 is designed to address.

The Fragmentation Risk

Before assessing which cooperation efforts generate returns under which AI futures, we must address a prior question: What is the cost of failing to cooperate at all? The United States has a self-interested reason to actively support EU and member state AI capabilities rather than to maximize its short-term leverage from EU dependence. The same imbalances that provide leverage today are simultaneously generating political pressure that will erode that advantage tomorrow, with EU entities increasingly inclined to build alternatives wherever feasible. A more capable partner is a more willing partner; structuring the relationship so that allied capability grows through cooperation is the most durable insurance against ecosystem fragmentation. That fragmentation—driven by eroding trust or an allied government’s conclusion that dependency has grown beyond what is domestically sustainable—could overlay any of the four scenarios described in the next chapter.

The trajectory toward this outcome is already visible. The U.S.-EU relationship is under considerable strain. Trade disputes, divergent threat perceptions about the Russia-Ukraine war, and a broader erosion of institutional trust have created an environment in which EU member state governments are becoming increasingly skeptical of arrangements that deepen their reliance on U.S. partners. That skepticism is neither irrational nor hostile; it reflects a legitimate desire to build indigenous capacity and exercise sovereign control over the infrastructure shaping EU member state economies. Importantly, a component of that pressure is political; it responds to perceptions of democratic legitimacy and strategic identity that favorable cooperation terms can moderate but not eliminate. The risk is a partnership reduced to its irreducible minimum: EU markets would still purchase chips and would still run workloads on U.S. cloud infrastructure because they must, but they would treat that dependency as a vulnerability to be managed rather than as a relationship to be deepened. In every domain where alternatives could feasibly be built, investment would flow toward building them.

A reasonable objection is that the structural barriers documented earlier in this analysis make it unlikely that the EU will achieve full technological autonomy within a policy-relevant time horizon, thereby weakening the urgency of the fragmentation threat. The objection is partially correct. A fully independent EU AI stack remains structurally out of reach under current conditions. But fragmentation does not require full autonomy to impose costs. Procurement preferences that disadvantage U.S. cloud providers in government contracts, data localization mandates, preferential regulatory treatment for EU-origin AI systems, and deliberate diversification toward non-U.S. partners in domains where alternatives exist are each achievable within EU institutional authority and are already visible in nascent form.

Since early 2025, dozens of European governments, municipalities, and public institutions have begun substituting U.S. providers for domestic or open-source alternatives. Most notably, the European Parliament switched its default search engine to France’s Qwant, the Dutch government

began migrating its code off of the Microsoft-owned GitHub, Finland declined to host its election data on Amazon's cloud, and cities across the Netherlands, France, and Germany are moving to replace Microsoft Office and Google Docs (Burgess, 2026).

These pressures are now migrating from practice into law. In June 2026, the European Commission proposed a Cloud AI Development Act designed to ensure that no provider of critical public workloads retains a "kill switch" over European systems, subjecting cloud vendors to sovereignty risk assessments on a four-tier scale and requiring that the hardware and software behind the most-sensitive public contracts be made in the EU (Nicol-Schwarz, 2026b; Chee, 2026).

None of these actions individually constitute a rupture, but their cumulative effect is a progressive narrowing of EU market access, data flows, and institutional integration on which U.S. AI firms' European revenue depends. The relevant threshold is not whether the EU can build a competing ecosystem in the next five years. It is whether the EU can impose sufficient friction on U.S. participation to erode the value of the transatlantic AI relationship incrementally over time. On that question, the structural barriers offer considerably less protection than they might appear to.

The cost of fragmentation extends beyond economics and diplomacy to the security infrastructure on which both blocs depend. The democratic bloc's capacity to monitor and respond to AI-enabled adversary activity depends on the same cooperation infrastructure whose erosion this section describes. Joint threat intelligence on adversary AI capabilities, coordinated red-teaming of models with national security implications, and shared methodologies for assessing AI-enabled influence operations all require the kind of trusted institutional channels that fragmentation progressively closes. A partnership that operates solely on transactional logic is also one in which the allied security architecture for AI threats operates with incomplete situational awareness.

The risks of a frayed relationship, moreover, are not confined to what the alliance forgoes; they extend to what a strategic competitor stands to gain. Given the high sovereign debt carried by many European states, a United States that lets the transatlantic relationship deteriorate risks creating the very conditions in which Chinese credit lines spread across the continent, purchasing goodwill and working to pry Europe loose from Washington (Juijn et al., undated).

Four AI Futures

The fragmentation dynamics described in the preceding chapter operate regardless of which trajectory materializes. The scenarios that follow address a different question: Under which technological futures do EU and member state capabilities generate the greatest strategic returns for the United States? The wide range of plausible outcomes can be meaningfully simplified into four archetypal scenarios—Concentrated Dominance, Commoditized Frontier, Dependent Specialization, and Collaborative Specialization. Looking toward 2035, the scenarios capture the range of outcomes that current technological trajectories and investment patterns could plausibly produce.

Two variables drive the analysis—the ultimate locus of value creation (general-purpose AI supremacy versus specialized models capture comparable or greater value) and model openness (closed-source dominance versus open-source remains competitive). These were selected over other candidate variables on two grounds. First, both remain contested among technical experts and industry analysts; scaling law evidence cuts in different directions on whether general-purpose models will continue to absorb specialized use cases, and the post-DeepSeek competitive landscape has left the closed-versus-open debate unresolved (Briski, 2025; Wolfe, 2025; Schmelzer, 2026; Edmond, 2025; Eastwood, 2026). Second, while other plausible drivers (including data abundance, computing power and infrastructure, investment levels, and algorithmic and model-architecture advances) independently shape AI development, their effects are largely legible through the two chosen variables, because they influence either where value concentrates or how accessible advanced models become. The reduction is not perfect.¹⁰

Table 4.1 summarizes both variables and several observable signposts that would indicate which trajectory is materializing.

¹⁰ A sufficiently radical architectural breakthrough, for instance, could simultaneously commoditize frontier capabilities and redistribute value across the stack in ways that do not map cleanly onto a single quadrant. The matrix is therefore best understood as a simplification that clarifies the policy-relevant strategic choices this analysis addresses rather than as an exhaustive representation of all possible AI futures. Its value lies in forcing conditional thinking across a range of structurally distinct outcomes, not in forecasting which outcome will arrive.

Table 4.1. Scenario-Driving Variables

Variable	Trajectory	Description	Comparative Signposts
Locus of value creation	General-purpose supremacy	Foundation models evolve to capably perform most economically important tasks, including in sectors where EU member states hold advantages.	• Revenue growth rates. Sustained outperformance by general-purpose AI signals a general-purpose trajectory. • Benchmark performance on comparable tasks. Higher general-purpose accuracy signals a general-purpose trajectory. • Rate of deployment. Faster general-purpose model uptake signals a general-purpose trajectory. • Rate of diffusion. Broader integration of general-purpose models signals a general-purpose trajectory.
	Specialized models capture comparable or greater value	Fine-tuned and narrow models capture comparable or greater value than general-purpose models. Domain expertise and proprietary data drive differentiation.	
Model openness	Closed-source dominance	Proprietary models pull decisively ahead of open-source alternatives, which yield substantially inferior performance on most economically valuable tasks.	• Benchmark performance differentials. A gap widening beyond 12–18 months of development signals closed-source models pulling away. • Rate of deployment. Enterprise-wide deployments concentrating around proprietary models signal closed-source dominance. • Rate of diffusion. Proprietary models crowding out open-source models signal closed-source dominance.
	Open-source remains competitive	Open-source alternatives remain close enough in performance benchmarks to the frontier to be leveraged for economic gain, even where they are not the best-performing systems available.	

Their intersection yields the following AI development scenario matrix (Table 4.2). Its four scenarios are analytical bounds, not predictions. Real-world development will likely combine elements of more than one. Their value lies not in forecasting correctly but in producing cooperation designs that prove approximately suitable—and correctable—across a range of plausible outcomes.

Table 4.2. AI Development Scenario Matrix

	Closed-Source Dominance	Open-Source Remains Competitive
General-Purpose AI Supremacy	Concentrated Dominance —A small number of firms would control proprietary models capable of performing most economically valuable tasks, including in several sectors where EU member states hold advantages. EU member state strategic offerings would narrow primarily to deployment and hardware-layer leverage through ASML, while asymmetric dependence would grow.	Commoditized Frontier —General-purpose models would dominate value creation, and no single provider could gatekeep access. Sectoral expertise would become less differentiating at the model layer, but it would retain value at the deployment layer, particularly in regulated domains. Robust governance would position EU entities as natural deployment partners for systems entering regulated markets.
Specialized Models Capture Comparable or Greater Value	Dependent Specialization —Fine-tuned and domain-specific models would capture a large share of economic value, rewarding EU sectoral expertise, but the proprietary foundation models on which most fine-tuning depends would be controlled by few firms. EU firms would operate from a position of dependency on U.S. model providers.	Collaborative Specialization —Open-source models would enable EU firms to leverage their deep sectoral expertise and established risk-management frameworks without as much structural dependency on providers. The EU would likely lead at the high-risk application layer.

Four assumptions underpin the scenario matrix. First, the analysis assumes that AI capabilities advance substantially over the next decade, whether through continued scaling, architectural breakthroughs, or a combination of both, such that AI systems become capable of performing a meaningfully broader range of economically and strategically valuable tasks than they do today. Second, the United States and China are assumed to continue leading in AI, and the scenarios do not adjudicate which country will prevail in the AI race. Third, ASML’s lithography monopoly is assumed to persist. Fourth, the scenarios assume that the EU and member states continue along the development path established earlier in the analysis, potentially narrowing, but likely not closing, the compute gap. EU member states have produced frontier-capable models, such as those by Mistral AI and Aleph Alpha, none of which yet match leading U.S. and Chinese models in performance, scale, or capitalization. That assumption is the starting point for assessment, not a verdict on European AI development potential.

Concentrated Dominance

Of the four scenarios, Concentrated Dominance typifies the most constrained operating environment for EU actors. In a scenario where a small number of firms control proprietary models capable of performing virtually every economically important task and open-source alternatives have fallen so far behind the frontier as to forfeit their utility as economic instruments, EU actors could face a dual squeeze.

First, foundation model providers could restrict access, impose punitive licensing terms, or deprioritize allied markets, leaving U.S. allies at the mercy of decisions made in boardrooms beyond the reach of democratic accountability. The resulting market structure—chokepoint control

concentrated in a handful of firms over a single critical access layer—would mirror the platform dynamics the EU’s Digital Markets Act was designed to constrain, yet no equivalent systematic remedy exists at the foundation model tier (European Commission, undated-a). Second, because general-purpose models could handle tasks that previously demanded deep sectoral expertise, the differentiation value of that expertise would erode, without disappearing entirely. Even the most-capable general-purpose models would likely still require adaptation for deployment into high-risk, regulated environments. EU and member state leverage would then narrow to two residual positions: regulated market access for firms seeking to scale proprietary systems across diverse EU environments and ASML’s lithography monopoly—the one asset in the EU’s strategic inventory whose value this scenario cannot diminish.

Whether the small number of firms controlling such models includes Chinese providers would materially affect the EU’s negotiating position, even absent realistic prospects for large-scale adoption. The EU’s security posture, GDPR constraints, and political resistance to Chinese infrastructure dependency make wholesale switching practically implausible. Yet credible alternatives need not be adopted to reshape bargaining dynamics. The existence of competitive Chinese models at or near parity may discipline U.S. providers’ pricing, licensing terms, and willingness to accommodate sovereignty requirements, because the prospect of selective adoption in lower-sensitivity commercial applications would become a latent but credible instrument for EU negotiators. That discipline would not rest on EU leverage alone. The commercial value of the European market gives U.S. firms their own stake in preserving access. Just as NVIDIA has lobbied Washington to keep selling chips to China rather than forfeit that market, model and AI infrastructure providers that depend on European revenue would have a commercial stake in offering guarantees that preserve it (Garicano and Grimm, 2026). It would be in the United States’ clear interest to ensure that U.S. model providers remain preferred not merely by dint of security constraints but also on the merits of the terms they offer.

The asymmetry inherent in this scenario, particularly if dominant providers chose to gatekeep access, impose extractive licensing arrangements, or redirect capacity toward other markets, could fundamentally destabilize U.S.-EU technological cooperation. Confronted with prospective technological marginalization, EU member states might pursue partnerships to build an independent AI ecosystem through coalitions with other middle powers. While such a coalition of democratic middle powers would present itself as nonthreatening, it would still attenuate U.S.-EU technological integration, generate duplicative development investment, and risk fracturing the standards coherence and interoperability essential for the democratic bloc to compete effectively against adversaries, weakening the broader alliance architecture and eroding the collective leverage that a unified democratic bloc commands against authoritarian competitors. The political dynamics that would animate such realignment, and the realistic forms it would assume, are developed in Chapter 3.

U.S. stakeholders would therefore benefit from structuring access to U.S. models on terms that preserve alliance cohesion. EU markets constitute a formidable venue for scaling proprietary systems across a heterogeneous landscape of regulatory environments, making private-sector investments in market entry and distribution partnerships essential for capturing the commercial and strategic value of EU member states’ sophisticated, densely regulated consumer and institutional bases. At the hardware layer, dedicated investment in ASML’s R&D would ensure that allied lithography capacity

continues to advance, preserving the allied position in the one domain where the EU holds a structurally irreplaceable advantage.

The primary mechanism through which that deployment-layer value would flow is via PETs and compliance infrastructure, technical instruments that enable computation on sensitive data without exposing the underlying information.¹¹ EU member state firms and institutions that have invested in transparency, auditability, and privacy-by-design techniques would be well positioned as natural providers of this infrastructure, particularly in sectors where GDPR and sectoral data rules create legal conditions that no contractual safeguard can substitute for (AI on Demand, 2024; European Commission, 2026c). Maturing the latter technologies is among the cooperation efforts addressed in Chapter 5.

Commoditized Frontier

Unlike Concentrated Dominance, this scenario dissolves the gatekeeper asymmetries that would most constrain EU actors; however, the advantages it confers are conditional, and their realization depends heavily on resolving the compute constraints discussed in the preceding chapters and on activating the governance qualities that, under this scenario, would function as a competitive asset. In a scenario where general-purpose models dominate value creation and their weights and technical documentation are circulated openly, EU firms and institutions could access frontier-capable models without the power asymmetries that define closed-source trajectories and build deployment solutions on top of freely available systems. Nonetheless, the EU and its member states would likely face compute constraints, and the value they could extract from this democratization of model access would depend critically on narrowing them.

For most AI applications, long-distance cloud compute would be technically sufficient, carrying no meaningful performance penalties. EU-controlled infrastructure, however, would remain preferable or, in some contexts, legally mandated for fine-tuning on sensitive workloads. Actors with greater access to affordable, abundant inference compute would therefore capture an outsize share of economic value. However, because general-purpose models could capably perform most economically important tasks, including in sectors where EU firms have historically held advantages, the returns to domain specialization at the model layer would contract, although a residual tier of sectors (i.e., those with sufficiently distinctive proprietary data inputs or demanding regulatory validation requirements) would retain niches where specialized development would still remain competitive.

When core technological innovation is freely available, economic returns would gravitate toward those who control the complementary assets required to deploy it profitably—what economists term the *complementary assets dynamic* (Hughes and Morton, 2006). In this scenario, those assets would include regulatory compliance infrastructure, sovereign compute, and sector-specific validation capabilities, precisely the areas where EU member states have made their deepest investments. EU

¹¹ The EU AI Act exempts open-source general-purpose AI models from certain transparency and documentation obligations to downstream providers (Articles 53–54), but these exemptions do not apply to models designated as having systemic risk or eliminate baseline copyright, training data, or GDPR requirements. Consequently, PETs remain relevant across the entire supply chain. Data controllers and processors can use PETs in training pipelines, foundation model providers can embed them in model-serving infrastructure, high-risk system deployers can build them into applications to reduce data protection impact, and third-party vendors can offer PET-backed assurance tools.

strategic value would therefore migrate accordingly, accruing less through model development and more through the capacity to deploy, validate, and certify AI systems in regulated environments. Whether the EU and its member states could fully capitalize on this complementary asset position would depend on how effectively they narrow the compute gap.

Under a Commoditized Frontier scenario, the EU's position improves because the competitive landscape reorganizes around the assets it already holds. The transatlantic relationship's strategic value would hinge on whether the EU can serve as the trust-and-compliance gateway through which openly available AI systems enter regulated markets globally. That value is real but conditional: It requires sufficient EU compute capacity and the diffusion of its governance frameworks, both trajectories the preceding analysis identifies as plausible and partially underway. One nontrivial consideration is security: Open-weight frontier-capable models would be available to adversary states and non-state actors alongside allied developers, elevating the salience of joint evaluation and responsible release coordination as indispensable pillars of alliance security.

Dependent Specialization

Of the four scenarios, this one positions the EU most ambiguously, because domain expertise acquires genuine market value but the dependency conditions under which that value must be realized blunt much of its strategic force. A scenario in which fine-tuned and domain-specific models capture a large share of economic value would vindicate the sectoral depth EU member states have accumulated. Still, the benefits would be unevenly realized if open-source alternatives have fallen significantly behind the frontier, leaving the foundation models on which much fine-tuning depends proprietary and controlled by a small number of firms. As in the Concentrated Dominance scenario, whether Chinese providers were among those holding proprietary foundation models would affect EU bargaining dynamics. Alternative non-U.S. providers, even imperfect ones, would expand the range of options available to EU firms seeking platform access. The landscape would likely bifurcate.

For applications built on fine-tuned closed-source models, EU firms would possess the domain knowledge, regulatory familiarity, and operational data that generate value. Nonetheless, they would build on platforms whose access terms they do not set. The resulting dynamic is what transaction cost economics terms a *hold-up problem*, whereby each round of fine-tuning investment binds an EU firm or organization more tightly to the specific model architecture on which it builds, raising switching costs and progressively weakening the firm's negotiating position at each contract renewal (Hart, 2007, p. 2).

The EU AI Act would partially offset this asymmetry, because high-risk AI providers operating in or deploying through EU markets would be obligated to furnish technical documentation, training data summaries, copyright policies, and instructions for use, giving EU firms mandated visibility into the models on which they build and an informational edge over developers in jurisdictions offering no equivalent transparency floor (EU Artificial Intelligence Act, undated-b). For applications where narrow models could be trained from scratch on proprietary operational data, EU firms would retain full autonomy over development and deployment, although the competitive standing of such models relative to those trained on leading frontier models would remain uncertain.

The resulting transatlantic dynamic would be mutually valuable but unevenly structured. Frontier providers would need regulated market access to deploy models into high-risk domains and would face constraints in sectors where models must be trained on datasets that reside exclusively within EU institutional contexts, including clinical data governed by national health systems, proprietary manufacturing sensor data, or financial compliance records subject to national regulatory regimes. These data advantages would provide EU firms with durable niches that frontier providers could not fill unilaterally, regardless of model capability. EU firms, however, would depend on proprietary foundation models for applications where building independently on open-source alternatives would yield significantly inferior performance. Hold-up logic suggests that the balance would tilt toward model providers, given that a frontier lab could seek domain partners across multiple jurisdictions, while EU entities' room to maneuver vis-à-vis proprietary model providers would be circumscribed by a far narrower set of alternatives.

That imbalance would not be static. If proprietary models continued to advance while open-source alternatives stagnated further, the EU and member states' negotiating position would likely erode incrementally, and the political constituency for alternative partnerships would likely grow, extending to non-U.S. providers or coalitions of middle powers. This dynamic reprises the fragmentation logic developed earlier, where asymmetry that generates near-term advantage simultaneously generates the political pressure that erodes it over time. How firms structure model access under these conditions is therefore among the most consequential decisions on this path.

A compounding risk runs through the supply chain dimension. If French financial compliance systems, German health care diagnostics, or Italian industrial automation come to depend on fine-tuning U.S. proprietary foundation models, the operational continuity of EU critical infrastructure would become contingent on decisions made by a small number of U.S. firms operating under U.S. legal and commercial logic. Disruptions at the model provider layer, whether driven by commercial restructuring, regulatory action, cyber incidents, or geopolitical calculation, would propagate into sectors whose continuity is a core security interest of the EU and member states. Mitigating this risk would require supply chain continuity provisions that effectively mandate private-sector behavior, an approach that I have deliberately excluded from the cooperation framework.¹² The risk nonetheless warrants identification as a vulnerability inherent to this scenario.

Collaborative Specialization

If the Concentrated Dominance scenario presents the most constrained environment for EU actors, Collaborative Specialization is the most generative, a scenario in which the conditions for transatlantic complementarity align most naturally, and the EU's accumulated sectoral depth translates most directly into shared strategic value. A scenario in which domain-specific, fine-tuned models capture value comparable to or greater than general-purpose AI models would most directly reward EU member state positioning, and the benefits would be most fully realized if open-source models remain competitive enough to serve as accessible foundations for that specialization. Under these conditions, EU firms could leverage freely available models alongside their deep sectoral

¹² This study focused on modes of cooperation between the United States, the EU, and its member states rather than on prescribing regulatory frameworks onto private-sector behavior.

expertise, proprietary operational data, and established risk-management frameworks to develop high-value applications without the access dependencies on proprietary model providers that define the closed-source scenarios. It is this combination, where specialization pays off and model access is open, that generates the broadest base of shared value.

The sectors where EU advantages would prove most consequential would be those requiring both deep institutional knowledge and robust compliance infrastructure, spanning industrial automation, health care and pharmaceutical research, financial compliance, and government services. EU firms would have two pathways to specialized value creation, each preserving meaningful autonomy: They could fine-tune openly available foundation models with domain expertise and proprietary data or build narrow models from scratch in domains where proprietary data are sufficient to train high-performing systems. Because open-source models could be fine-tuned freely, EU firms would not need to negotiate access terms or licensing arrangements to build on frontier capabilities. EU governance frameworks would further consolidate the EU's position by ensuring the responsible development of precisely the high-risk applications where specialized models would command global premiums.

Under these conditions, the transatlantic relationship would reach its point of greatest natural complementarity. The underlying market logic is one of comparative advantage, with the United States holding relative efficiency at foundation model advancement and EU member states holding it at domain application and regulatory validation—both capturing more value by trading along those lines than by pursuing autarky across the full stack. That division of labor is not merely theoretical; as the sectoral applications surveyed earlier demonstrate, it already characterizes the most productive transatlantic AI partnerships in the ecosystem. The access constraints that weigh on actors across the other three scenarios would be largely absent, replaced by a self-enforcing division of labor whose terms neither party needs to be coerced into accepting. The deciding factors on realizing that complementarity would be EU sovereign compute, whether specialized EU firms can reach global markets at scale, and the continued maturation of the open-source ecosystem on which their relative autonomy depends.

Structuring Cooperation Under Uncertainty

Transatlantic AI cooperation cannot be optimized in advance; it benefits from being designed to evolve. The AI future scenarios examined in Chapter 4 demonstrate why. The strategic value of EU and member state capabilities is not fixed; it shifts materially depending on whether AI value concentrates in proprietary foundation models or distributes across specialized applications, as well as on whether open-source alternatives remain competitive or fall decisively behind the frontier. The planning challenge this creates is not unique to AI: Colin Gray's work on defense planning under conditions of irreducible uncertainty counsels that when the future is "unknown and unknowable in detail," the goal is not to predict correctly but to build architectures tolerant of error, adaptable to emerging conditions, and disciplined by political reality rather than analytical elegance (Gray, 2019, p. 173). A cooperation framework calibrated to any single assessment of that value risks being well positioned for one future while poorly structured for several others.

That does not imply that existing cooperation mechanisms have failed. Bilateral frameworks are functioning, private capital is already reshaping EU and member state infrastructure at a scale that no government program could match, and both public and private actors are engaged in forms of coordination that approximate many of the priorities identified in this analysis. The argument advanced here is that approximation is insufficient when the strategic value of cooperation is itself scenario-dependent. What the existing architecture has not produced is a conditional logic that signals to policymakers and industry leaders which priorities might rise and which might recede, as observable signals indicate the technological trajectory materializing.

The initiatives that follow are not exhaustive; a comprehensive screening of current and potentially worthwhile cooperation efforts is outside the scope of this analysis. The five cooperation efforts offered below, identified through the study's capability- and scenario-based analysis, instead serve as a menu of engagement options across different technological futures. Many of the investments described here have analogs in current practice. Where existing efforts are already underway, this analysis identifies what additional structuring or scaling could help achieve stronger outcomes. Where the initiatives are more novel, that distinction is noted explicitly. Where specific mechanisms, institutional arrangements, and funding structures are identified, they are offered as illustrations of what implementation could look like rather than as prescriptive recommendations. The determination of specific instruments, mechanisms, and institutional arrangements through which these options are pursued is appropriately left to the implementing actors best positioned to assess the political, legal, and operational constraints of their specific contexts.

The preceding chapters establish that geopolitical fragmentation is a cross-cutting risk that operates independently of which technological scenario materializes, driven by the erosion of trust, the

accumulation of dependency, and the political pressure those dynamics generate over time. Each cooperation effort in this chapter has been selected, in part, because its pursuit, particularly when responsive to the signposts identified in the preceding chapter, could help counter that risk.

Table 5.1 provides an orienting overview of the five cooperation efforts and their applicability across the four scenarios described in the preceding chapter. Baseline efforts—those scoring high across all four scenarios—generate strategic returns regardless of which trajectory prevails and could be understood as the most foundational priorities given that their returns hold across the full range of plausible outcomes. *High* priority indicates that the investment is among the most consequential cooperation levers under that scenario. *Moderate* priority indicates that the investment retains meaningful value but is not the primary lever. *Low* priority indicates that the investment generates limited returns under that scenario and would tend to offer less value than higher-priority alternatives competing for the same resources. The initiative descriptions that follow provide the analytical foundation for these assessments.

Table 5.1. Low-Regret Investments, by AI Future Scenario

Cooperation Effort	Concentrated Dominance	Commoditized Frontier	Dependent Specialization	Collaborative Specialization
Evaluation, testing, and standards interoperability	High	High	High	High
Lithography resilience	High	High	High	High
Compute access for the broader EU innovation base	Low	High	Moderate	High
PET joint collaboration	High	Moderate	High	Moderate
Sectoral commercialization partnerships	Low	Moderate	High	High

NOTE: Priority levels are as follows: high = primary cooperation lever; moderate = meaningful but secondary; and low = limited returns, should not crowd out higher-priority efforts. *Low-regret investments* denotes cooperation efforts that generate strategic returns without requiring accurate prediction of which AI development scenario will materialize. Drawing on the counsel of Gray (2019) that, under irreducible uncertainty, strategy should prioritize architectures tolerant of error, the term remains relevant to both tiers of cooperation developed in this chapter: (1) *foundational* efforts, whose returns are scenario-invariant and therefore warrant immediate pursuit, and (2) *contingent* efforts, whose returns are scenario-sensitive but remain meaningful across a sufficient range of futures to justify building institutional readiness for them now.

Interpreting the AI Development Scenario Matrix

The ratings in Table 5.1 reflect a scenario-by-scenario assessment of each effort, examining the mechanism through which it generates returns under a given trajectory and the conditions under which that mechanism would weaken. Two groupings emerge from that assessment and organize the discussion that follows: a *foundational* tier, whose returns hold across all four scenarios, and a *contingent* tier, whose returns vary with which trajectory materializes.

The foundational tier comprises the two efforts scoring high across all four scenarios. Each addresses a strategic asset or function whose value to the alliance is preserved across divergent

technological futures. Lithography illustrates the pattern at its cleanest; frontier model development continues to demand the most-advanced chips across every scenario examined, regardless of where value is concentrated. The evaluation, testing, and standards interoperability effort holds value for a related but distinct reason; under every scenario, the alliance benefits from shared reference points for assessing AI systems, although the specific risks those reference points address shift with the technological environment. In both cases, the asset or function is sufficiently load-bearing such that no plausible scenario can eliminate its strategic importance, even when the precise theory of value differs.

The contingent tier operates on a different logic. The three efforts in this tier address constraints whose strategic salience is itself scenario-sensitive. Compute access for smaller EU actors matters most when open-source models remain competitive, because that is when compute, rather than model access, becomes the binding constraint on participation. PETs matter most when sensitive data must leave EU institutional environments to reach externally controlled models, a condition most pronounced under the proprietary model trajectories. Commercialization partnerships matter most when specialized applications capture meaningful economic value—a condition that general-purpose scenarios partially erode. Ratings across the contingent tier, therefore, vary because the underlying problem each effort addresses varies in salience with the scenario.

Foundational Efforts

Both of the following cooperation efforts score high in Table 5.1 because they generate returns regardless of the scenario. Policymakers and industry leaders who disagree on which scenario is most likely may nonetheless find common ground here.

Evaluation, Testing, and Standards Interoperability

The ability of U.S. and EU actors to jointly evaluate AI systems, compare results across languages and deployment contexts, and maintain common safety standards is the connective tissue of the democratic AI ecosystem. Without it, the transatlantic relationship lacks the shared reference points that make coordinated governance possible and that prevent a standards divergence from leading to self-fulfilling fragmentation. This is why the effort scores high across all four scenarios in Table 5.1.

The institutional foundation for this effort exists but is somewhat dormant. The 2022 U.S.-EU Trade and Technology Council (TTC) Joint Roadmap suggested activities to establish shared terminologies and taxonomies, collaborate on international technical standards and tools, and monitor existing and emerging AI risks to ensure interoperable risk management (European Commission, 2022b, p. 2). By April 2024, the TTC had also developed two lists of key AI terms with mutually accepted definitions (EU-U.S. Trade and Technology Council Working Group 1 Subgroup on AI Taxonomy and Terminology, 2024). In practice, the TTC's working group architecture, paired with its ministerial cadence, facilitated meaningful procedural and technical coordination. It convened the relevant agencies on both sides, set shared work programs, produced jointly authored deliverables, and provided a recurring forum for reviewing and advancing AI-specific outputs. Technical dialogue between the European AI office and the U.S. AI Safety Institute has also explored watermarking, government compute infrastructure, and AI-for-good use cases (European Commission, 2024a).

That architecture, however, has not convened at the ministerial level since 2024, and its future remains uncertain. The current political conditions may not be favorable to reaffirming this type of transatlantic relationship.

Three partial substitutes remain operational and matter for what follows: the International Network of AI Safety Institutes, the bilateral technical relationship between the United States' National Institute of Standards and Technology (NIST) and the European AI Office, and the OECD's AI principles (within the broader OECD Council on Artificial Intelligence recommendation), to which both blocs adhere (Bengio et al., 2026; NIST, 2024; Executive Office of the President, 2025, p. 22; OECD, undated-c; Dunietz et al., 2024, p. 3). Each provides coordination capacity that is independent of the TTC's ministerial format, although none replaces its convening function.

What the current architecture has not yet produced, and what relative dormancy on the subject has made more urgent, is the institutional permanence and operational depth that alliance-grade coordination would call for. The existing stack is better described as a network of institutions whose connective tissue has frayed than as a fully resourced, durable structure aligning U.S. and EU governance practices.

Two gaps are particularly consequential. First, the function performed by the TTC's working groups and ministerial meetings has not been replaced by an equivalent mechanism, and the alternatives currently in operation are valuable but episodic, depending on sustained political will from institutional leadership rather than on a dedicated mandate that endures across administrations.

Second, there exists no reciprocal arrangement for models in high-risk domains comparable to the bilateral agreements that the United States' Center for AI Standards and Innovation (CAISI) has developed with specific U.S. firms, meaning that AI systems entering regulated sectors are not yet streamlined through a joint transatlantic safety assurance process (Bordelon, 2026). The absence of such an arrangement may grow particularly consequential as national security considerations rise on the pre-deployment agenda. Catalyzed by concerns raised by Anthropic's new Mythos model, the United States has begun weighing whether AI models should be vetted ahead of release, a deliberation whose outcome, in the absence of a transatlantic counterpart process, may leave allied regulators evaluating the same systems against asymmetric criteria and on asymmetric timelines (Bordelon, 2026).

Closing those gaps could involve organizing existing institutions into a more durable form. However, the political conditions for new permanent institutions do not currently exist. A productive framing would distinguish between what is achievable within existing institutional mandates and what could be designed now for implementation when political conditions permit.

In the near term (i.e., within the next year or two), what could be pursued would operate through proven durable channels—the International Network of AI Safety Institutes, the NIST–European AI Office bilateral technical relationship, and the OECD AI principles. Two categories of deepening engagement are achievable without new political authorization. One possibility is more frequent convenings with standing agenda items and coordinated red-teaming exercises that build the operational habits that a more formal structure would eventually benefit from. A second is substantive pilots, including shared test protocols for evaluating high-risk systems and mutual recognition arrangements for conformity assessment in specific sectors, beginning where regulatory frameworks

are already most closely aligned. These are not substitutes for institutional permanence, but they are the activities on which permanence could credibly be built without depending on available political will.

On a longer horizon, what could usefully be designed now, for implementation when conditions permit, is the institutional form that would make such coordination self-sustaining rather than dependent on the continued initiative of individual agency leaders. A permanent joint secretariat would address that requirement by providing a continuous brokerage function that rotating institutional leadership and episodic ministerial engagement cannot reliably sustain. To be politically viable when its moment arrives, such a secretariat would likely need to be lighter and more technical than the TTC's ministerial-led format, anchored in the agency-to-agency relationships described above.

Equally important is how the secretariat is framed. The TTC's governance-first framing proved vulnerable to the shift in political conditions that followed its creation. A secretariat framed instead as serving the geopolitical competitiveness and national security interests that both blocs' leadership prioritizes would carry less political baggage across future administrations. Once established and functioning, such a secretariat could pursue more-ambitious coordination goals, including reciprocal pre-deployment access arrangements for AI systems entering high-risk sectors comparable to the bilateral agreements that CAISI has developed with specific U.S. firms—closing the second gap identified above.

Lithography Resilience

As discussed in the preceding analysis, ASML's High-NA EUV monopoly makes it one of the most consequential hardware chokepoints in the allied AI ecosystem, and its continued capacity to innovate under increasing geopolitical pressure constitutes a shared strategic interest under every AI trajectory. Frontier model development continues to demand the most-advanced chips, regardless of whether AI value is concentrated in general-purpose or specialized models and regardless of whether open-source alternatives remain competitive, which is why this effort, like the evaluation, testing, and standards interoperability effort, scores high across all four scenarios in Table 5.1. Any disruption to R&D continuity, supply chain integrity, or operational security of the allied lithography base would constitute a crisis.

Meaningful cooperation already exists across both governmental and research layers of the lithography supply chain. On the governmental side, the most substantive efforts have developed bilaterally between the United States and the Netherlands, driven primarily by export control alignment (Bermingham, 2026). On the research side, U.S. entities maintain active R&D partnerships with Imec in Leuven, Belgium, which serves as ASML's de facto external R&D ecosystem (Bologa, 2025; Imec, undated-b).

Three gaps are particularly consequential. The first is an R&D funding gap. Coordinated long-horizon R&D funding commitments to sustain High-NA EUV innovation remain largely absent, leaving ASML's development pipeline exposed to the revenue pressure that tightening export restrictions has begun to impose, particularly given that ASML relies heavily on its own revenue for R&D (Kharpal, 2026; ASML, 2025a, pp. 32, 55; Norges Bank Investment Management, 2025,

minutes 7–9, 13). The second is an emergency allocation gap. There exists no publicly available information about a pre-agreed emergency allocation arrangement with the U.S. government for crisis conditions. In its absence, the distribution of ASML’s output during a supply crisis would be governed solely by commercial interests. The third gap concerns the publicly documented architecture for cyber and operational security. Cyber activity against ASML, such as the 2015 breach of its IT systems, may become more frequent in the current competitive environment (ASML, 2025b). Beyond the EU Chips Act’s monitoring and alert provisions, the open record does not surface an ASML-specific allied framework for joint threat intelligence-sharing or coordinated incident response (European Commission, 2026d). Classified arrangements may exist beyond the open-source purview of this study; even so, a publicly visible framework carries deterrence and alliance cohesion value that nonpublic coordination cannot fully replicate.

Closing such gaps could involve more-binding commitments across three dimensions. **On R&D funding**, a formal U.S.-EU-Netherlands semiconductor board is one option that could help ensure that allied investment priorities remain aligned with the pace of innovation demanded by the AI ecosystem. The board’s design would need to reckon with the fact that EU member state–aid rules constrain direct R&D subsidy in ways the U.S. legal framework does not, which is why Project Beethoven invests in output capacity rather than in R&D directly. Rather than treating such constraints as an obstacle, the board’s architecture could be structured around them, coordinating across three funding channels that respect each party’s legal constraints:

- *One possible channel is direct U.S. R&D commitments through existing CHIPS Act and Department of War mechanisms,¹³ which face fewer constraints on long-horizon support than their EU counterparts. Multiyear commitments structured to extend across project cycles rather than annual appropriations could help insulate ASML’s development pipeline from the revenue pressure that tightening export restrictions has begun to impose.*
- *A second channel is additional EU contribution through the Important Projects of Common European Interest (IPCEI) designation, the same legal vehicle that anchors the European Chips Act. IPCEI can authorize public funding for R&D, but it requires multilateral review to establish common European interest and proportionality, which builds in deliberation that U.S. channels do not face (European Commission, undated-e). A revamped EU Chips Act, proposed in June 2026, may narrow that gap by permitting direct EU investment in strategic cross-border projects, a provision that could lower procedural barriers to financing ASML’s development pipeline (Chee, 2026; “Europe’s Tightrope Walk on Tech Sovereignty,” 2026). Even so, structuring the board’s coordination function to account for residual timeline differences is what would allow commitments to compound.*
- *A third channel is a renewed customer co-investment program modeled on ASML’s 2012 arrangement, which sat entirely on the commercial side of EU member state–aid rules (ASML, 2012a). The competitive landscape that produced the original program has only*

¹³ The Department of War is designated the Department of Defense under Public Law 81-216, National Security Act Amendments of 1949. See Public Law 117-167, CHIPS Act of 2022, August 9, 2022.

intensified with AI demand, and the board could facilitate a new generation of such commitments.

Across these three channels, the board's core functions could include joint capital roadmap reviews tied to ASML's published technology trajectory, coordination of timing and scale across the channels, and a mechanism for assessing the revenue impact of incremental export restrictions before they are imposed, closing the loop between the security architecture that constrains ASML's market and the innovation architecture that depends on it.

On emergency allocation, a preset supply prioritization arrangement offers one avenue to address the second gap. Such an arrangement would not need to specify allocation outcomes in advance; it could more usefully establish the consultation mechanism, decision criteria, and institutional authorities that would govern distribution during a supply crisis so that the framework exists before it is needed rather than being negotiated under duress. The appropriate venue for such negotiations depends on which institutional channels are available when political conditions permit. The TTC, were it to reactivate, would be one candidate.

On cyber and operational security, building on the U.S.-Netherlands Cyber Dialogue and the U.S. Department of Homeland Security's relationship with its Dutch counterparts, the United States could consider extending an ASML-specific framework for joint threat intelligence-sharing and coordinated incident response, particularly as ASML's facilities continue to expand globally (U.S. Department of State, 2023; Stegon, 2012; ASML, undated-b). To capture the deterrence value, such a framework would benefit from some degree of public visibility, drawing on models of arrangements for other categories of critical allied infrastructure and recognizing that ASML's strategic position warrants comparable protection (Cybersecurity and Infrastructure Security Agency, undated-a; Cybersecurity and Infrastructure Security Agency, undated-b).

Scenario-Contingent Priorities

The cooperation efforts in this section reward attention to the trajectory. Each becomes consequential depending on which scenario materializes, as described in the "Conditional Logic" subsection for each effort below. The policymaker or industry leader who monitors the signposts identified in the preceding chapter will find in them a set of priorities that sharpen as the AI development landscape clarifies.

Compute Access for the Broader EU Innovation Base

As discussed in the preceding analysis, sufficient sovereign compute is a meaningful enabler of EU AI development, and its accessibility shapes which actors can participate in that development and at what scale. *This matters most under scenarios in which open-source models remain competitive, because it is under those conditions that the binding constraint shifts from securing access to capable models, which open weights largely resolve, to securing the compute needed to fine-tune and serve those models independently.*

Substantial U.S. private capital is already reshaping EU compute infrastructure. Amazon's almost €34 billion Spanish cloud expansion exemplifies a much broader pattern of hyperscaler investment that is materially narrowing the EU's compute gap (Latona and Pinedo, 2026). Public infrastructure is

also expanding. Multiple EuroHPC JU AI gigafactories have come online and explicitly prioritize SMEs and startup access (EuroHPC JU, 2025; EuroHPC JU, 2026).

Two gaps nonetheless remain consequential. The first is an allocation gap. Hyperscaler-anchored capacity is often absorbed before it reaches the broader market; Start Campus's SIN01 facility, though technically constituting EU sovereign cloud, was fully leased to Microsoft before operations began (Dunn, 2025). The pattern concentrates compute among large enterprise customers and underserves SMEs, startups, and research institutions where specialized innovation also occurs. This gap operates directly on the Capital → Compute interaction identified in Figure 2.1. Capital scarcity among smaller EU actors prevents them from securing the compute required for participation in the AI ecosystem. It also bears on the Compute → Talent interaction. Where compute access is scarce, researchers who depend on it must follow it elsewhere. The second is a coverage gap. EuroHPC JU's gigafactories may not meet aggregate demand from the smaller actor base on their own, particularly as that demand grows with the maturation of open-source models.

Closing these gaps could involve structuring U.S. private investment such that a portion of hyperscaler capacity reaches the smaller actors that market allocation patterns currently underserve.

On allocation, one option would be for the U.S. government to use foreign policy vehicles—including the American AI Exports Program—to incentivize hyperscalers to reserve a portion of their EU compute allocation for SMEs, startups, and research institutions with high potential to produce novel breakthroughs (American AI Exports Program, undated). Available tools include streamlined export licensing for providers that meet access standards and preferential treatment in U.S. and EU government procurement frameworks. Hyperscalers do not normally need an export license to run cloud infrastructure serving foreign clients, but certain AI- and compute-related exports, such as advanced chips and high-capability AI models, are subject to such controls, and these controls are increasingly shaping the terms on which that infrastructure is provided (Heim, 2025a). That regulatory surface is what makes foreign policy instruments a viable lever. It gives the U.S. government meaningful incentives to offer without mandating private-sector behavior.

On security vetting, any access program would benefit from incorporating calibration to the EU member state in which the recipient is established. Adversary technology exposure is unevenly distributed across European geography, resulting in varying vulnerability profiles with implications for access structure. China's manufacturing presence in Hungary, China's equity stake in Portugal's primary electric utility, and Huawei's operation of AI research centers in Germany, for instance, may each bear on trusted-partner determinations in those specific national contexts (O'Carroll and Garamvolgyi, 2026; Bugge, 2011; Huawei Research Center Germany, undated). Preferential compute access extended to a recipient in a sector with documented adversary-linked equity or infrastructure generates supply chain risks that the program design would do well to anticipate. Conditioning access on effective member state screening regimes or on recipient-level vetting could ensure that the access program accounts for such variability.¹⁴

¹⁴ The access-tiering logic underlying this vetting approach could mirror that established by the AI Diffusion Rule, which sought to calibrate compute export permissions to recipient-country risk profiles and end-user vetting standards. Public statements—including those from the Executive Office—have characterized this rule as revoked; however, as of the time of writing, it remains legally in force, although it is not being actively enforced (Bureau of Industry and Security, 2025). Its underlying architecture nonetheless offers a relevant regulatory precedent for the kind of geography- and recipient-sensitive access structuring recommended here.

A reasonable objection is that channeling hyperscaler capacity toward smaller EU actors deepens the very dependency on U.S. cloud infrastructure that EU sovereignty ambitions seek to reduce. The objection has merit, but three considerations weigh against it. First, the counterfactual is not having EU sovereign compute fill the gap; EuroHPC JU may not meet aggregate demand on its own, and without structured access, hyperscaler capacity may continue to concentrate among large enterprise customers while smaller actors remain underserved. The dependency exists either way; the choice is whether it broadens access. Second, firms that rely on hyperscaler infrastructure are nonetheless EU firms whose development strengthens the indigenous innovation base. Third, the arrangement is reversible as EU sovereign compute matures.

Conditional Logic

The returns for such incentivization would be highest under trajectories in which open-source models remain competitive, because it is under those conditions that compute access would become the primary determinant of which actors can participate meaningfully in the AI ecosystem. Under the Commoditized Frontier and Collaborative Specialization scenarios, EU member state firms and institutions with sufficient sovereignty-compatible compute could leverage freely available models to develop the high-risk applications where EU member states hold their strongest competitive advantages and where the combination of open-source model access and sovereign compute would create the most-favorable conditions for specialized value creation.

Under the Dependent Specialization scenario, compute access would retain meaningful but secondary value. For most AI applications, long-distance cloud computing is technically sufficient, and in-region sovereign compute would become a binding requirement only when regulations mandate or encourage domestically controlled infrastructure. For instance, a U.S. firm that has successfully navigated the EU AI Act's transparency and review obligations may still find that a model or application cannot be deployed on U.S.-hosted cloud infrastructure if the data involved are subject to sovereignty requirements. In-region compute, therefore, would become the final enabling layer for U.S. firms seeking to further monetize their EU regulatory compliance investments.

How consequential that layer proves under the Concentrated Dominance scenario depends on how commercially significant the high-risk application developer base becomes relative to the broader enterprise market. If regulated sectors come to represent a meaningful share of U.S. providers' revenue ambitions (and those are precisely the use cases where closed-source frontier models fall short), sovereign compute access for smaller EU developers would become salient.

PET Joint Collaboration

Deploying AI in regulated sectors depends on technical mechanisms that make trustworthy deployment demonstrable to regulators, partners, and public actors whose data such systems process. PETs address that function by enabling AI development and deployment of sensitive data without exposing the underlying information (OECD, undated-b). *Their strategic salience is greatest under proprietary model trajectories, where closed-source architectures draw sensitive data outward to externally controlled infrastructure.*

The centrality of PETs is not solely a matter of best practice; it is a legal condition of AI deployment in some of the EU's most valuable sectors. GDPR's data minimization and purpose limitation principles, combined with sectoral rules governing medical records, financial data, and government personal information, create conditions under which sensitive data cannot be exposed to a model during training or inference, regardless of what contractual or procedural safeguards are in place (Intersoft Consulting, undated). Only technologies that enable computation on data *without* exposing their underlying information permit lawful AI development in these domains.

Two gaps are consequential. First, despite the centrality of this technical layer, no publicly available information points to a dedicated transatlantic cooperation pillar on PETs currently in place, even as both the United States and the EU have expressed considerable interest in maturing these technologies (NIST, undated; European Commission, 2023a). That such a pillar is achievable is demonstrated by the U.S.-UK PETs prize challenges, which established that bilaterally structured, regulatory-linked competition mechanisms can produce deployable, privacy-preserving AI solutions (NIST, 2025). Second, the demand for PETs across these layers is clear, but the supply is not yet commensurate. Fully homomorphic encryption, secure multi-party computation, and general zero-knowledge systems have improved substantially but often remain impractical for large-scale or low-latency workloads (Wiewiórowski, 2025; Zewe, 2025; Oguro, undated). These maturity gaps are precisely what a dedicated cooperation pillar could target.

Cooperation in this domain could therefore operate on the Data → Capital edge. Investors pricing AI deals in regulated sectors tend to account for the legal uncertainty surrounding sensitive data use—uncertainty that mature, standardized PETs would materially reduce.

For research coordination, one immediately actionable path would be a dedicated PETs cooperation dialogue stream between NIST and the European AI Office. The resulting outputs could provide the governing framework for jointly funded research into privacy-preserving fine-tuning and secure inference, building on the U.S.-UK PETs prize challenge model that has already demonstrated bilateral viability. The EU's testing and experimentation facilities could offer the sectoral infrastructure for validating jointly developed solutions in real-world deployment contexts, bridging the gap between research output and regulator-validated implementations that can be deployed at scale.

Such a dialogue would need to address multiple points in the AI supply chain, because privacy constraints bind at each stage where sensitive data are processed. In data pipelines and training workflows, PETs enable model development using datasets that cannot be legally exposed in raw form. In model-serving infrastructure and secure compute services, they permit inference on protected inputs without requiring data to leave controlled environments. In high-risk applications, they enable deployment that meets both the EU AI Act and sectoral requirements. In assurance tooling, they help demonstrate compliance to regulators by providing auditable, privacy-preserving records of system behavior. A cooperation pillar that addresses only one layer would leave the others as points of failure, causing deployment to stall regardless of progress elsewhere.

Private co-investment would be valuable to cover all these layers. U.S. cloud providers and AI developers operating in regulated EU markets have a commercial interest in the maturation of PETs, because demonstrating trustworthy AI deployment is increasingly subject to market access; their

participation as co-funders and implementation partners could accelerate the transition from research investment to a deployable, scalable component of the stack.

Conditional Logic

Returns to PET investment would be highest under scenarios in which proprietary models dominate, because it is under those conditions that the exposure surface that PETs are designed to protect is at its widest.

Under the Concentrated Dominance and Dependent Specialization scenarios, closed-source architectures would require that training and inference workloads execute on infrastructure controlled by a small number of external providers. The “Abundant Data, Restricted Flow” section in Chapter 2 establishes that, for instance, clinical records are governed by GDPR and sectoral rules and cannot be lawfully transmitted to such providers in raw form. PETs would supply the technical custody layer that would permit regulated-sector deployment on proprietary models without surrendering control of the underlying data. The narrower the pool of providers and the more concentrated the potential points of exposure, the higher the stakes for that custody layer functioning reliably.

Compounding that concern, the opacity of closed-source models means that EU actors cannot independently verify how their data are processed once they cross into provider-controlled infrastructure, making cryptographic guarantees a substitute for the inspection that proprietary licensing forecloses.

Under the Commoditized Frontier and Collaborative Specialization scenarios, the exposure surface would somewhat contract. Open-weight models could be deployed on-premises, keeping sensitive data within the boundaries that generate and govern them. Fine-tuning on proprietary operational data could proceed within the developer’s own compute environment, mitigating cross-boundary data flows that drive PETs’ centrality in proprietary model scenarios. PETs would retain value under these trajectories through federated learning across institutions and cross-border collaborations, which require techniques that on-premises deployment cannot provide. Nonetheless, the absolute returns would be lower than under trajectories where data must leave the institution to reach the model in the first place.

Sectoral Commercialization Partnerships

Deep sectoral expertise, proprietary operational data, and established regulatory sophistication together constitute genuine competitive advantages for EU member states, but the financing and distribution infrastructure that would systematically convert those advantages into globally scaled products remains underdeveloped relative to U.S. counterparts. *This matters most under trajectories where specialized applications capture meaningful economic value, because it is under those conditions that the commercialization bottleneck becomes a binding constraint on whether EU expertise generates increased returns at scale.*

As the “Distributed Excellence in AI Applications” section in Chapter 2 documents, transatlantic commercialization partnerships for specialized AI already exist and have worked extraordinarily well. Of the EU member state firms shown in Table 2.2, 15 of 16 maintain active U.S. partnerships spanning investment, distribution, cloud infrastructure, and research collaboration. Mistral AI’s reach

into global markets through U.S. cloud distribution channels and funding, while retaining sovereignty over sensitive deployments, illustrates this model at its most developed.

Two gaps nonetheless persist. The first is the financing gap. EU specialized AI firms consistently lack access to growth and late-stage capital at the scale that would bring their products to global markets. The second is the go-to-market resourcing gap. Beyond capital, smaller specialized firms may lack the distribution relationships, cloud and platform integrations, and commercial infrastructure needed to reach U.S. and global markets. The Mistral AI model has worked, but it has worked for firms that are already large, visible, and well-resourced enough to negotiate these arrangements independently; the broader population of EU specialized AI firms absorbs transaction costs that make equivalent partnerships less accessible at smaller scales. Both gaps operate on the Capital $\rightarrow$ Talent interaction identified in Figure 2.1: The absence of scaling pathways drives talent outflow, which in turn compresses the funder pipeline that shapes future investor confidence.

Supporting the broader offerings of the EU specialized AI ecosystem could be approached through a single bilateral mechanism that performs two functions: pooling and coordinating U.S. capital toward specialized EU firms and brokering their access to the broader commercialization infrastructure required to compete globally.

On financing, a U.S.-EU Strategic Technology Investors Council (STIC), modeled loosely after the U.S.-UK precedent, could function as a coordinated co-investment platform anchored by participating U.S. venture firms and institutional investors. This council could be co-led by the U.S. International Development Finance Corporation, the U.S. Department of Commerce, and their EIB and EU counterparts (The American Presidency Project, 2024). The U.S.-EU STIC's core functions could include scouting EU firms with high-value prospects; facilitating AI-relevant funding and deal flow across health care, manufacturing, robotics, and other strategic sectors; pre-vetting EU firms against security and regulatory criteria; and structuring co-investment terms that lower the diligence costs that U.S. investors may bear when entering EU markets. The objective would not be to subsidize firms but to reduce the friction that has kept U.S. capital concentrated among the most visible companies.

On go-to-market resourcing, the STIC could connect vetted EU firms with U.S. sovereign cloud providers and AI platform operators, helping absorb some of the networking costs that currently make such partnerships less accessible to smaller actors. The aim would be to replicate, for the broader specialized AI firm population, the commercialization advantages that the largest EU firms have secured independently by providing them with STIC endorsement. Such a setup could allow a more politically digestible framing on both sides. On the U.S. side, a Department of Commerce–led council is more readily positioned to secure U.S. access to high-value EU sectoral expertise. On the EU side, viability depends on the STIC facilitating capital flows and commercialization infrastructure that EU-native sources are more risk-averse to provide at scale, rather than displacing the native capital already flowing.

On security vetting, participation would benefit from being calibrated to the EU member state in which the recipient is established. As discussed in previous efforts, adversary technology exposure is unevenly distributed across European geography, and the same calibration logic applies here. Conditioning STIC participation on effective member state screening regimes or on vetting at the recipient level could ensure that the program accounts for such variability.

A reasonable objection to this effort would be that systematically channeling U.S. capital and distribution access toward EU firms further entrenches the dependency that EU sovereign AI ambitions seek to reduce. The objection has merit: A STIC that succeeds in its core function will, by default, deepen the integration of EU firms into U.S. capital and distribution networks. The countervailing consideration is that the alternative is materially worse. More technology startups are founded in Europe than in the United States, but they routinely fail, relocate, or fail to scale (Schörner, 2026). The Scaleup Europe Fund and analogous instruments should help over time but may not close the gap that the U.S. ecosystem is well equipped to fill (European Innovation Council, undated). The marginal sovereignty cost of deeper U.S. engagement is lower than the capability losses its absence could produce.

Conditional Logic

The returns on such an effort would be highest in scenarios where specialized applications capture meaningful economic value and more limited where general-purpose models come to erode the market for domain-specific expertise.

Under the Dependent Specialization and Collaborative Specialization scenarios, the commercialization bottleneck would be a binding constraint for EU specialized AI firms. What would prevent EU expertise, data, and governance advantages from generating returns at scale would be the structural lack of financing and distribution connections required to bring specialized products to global markets. Closing that gap would be among the most consequential priorities for cooperation under either trajectory.

Under the Commoditized Frontier scenario, the returns would be meaningful but secondary. General-purpose models would dominate value creation, shifting competition primarily to the deployment layer. More-mature EU firms would be better positioned to capture deployment-layer value than those without it, but strengthening EU firms' sovereign compute access and PET infrastructure would be the more urgent cooperation priorities under this trajectory.

Under the Concentrated Dominance scenario, the logic would shift materially. If general-purpose models can effectively replace most domain-specific ones, the addressable market for EU specialized AI applications shrinks. Apart from comparative niches that will likely appear, returns would be limited enough under this trajectory that they would not warrant displacing the foundational efforts.

The Window and Its Closure

The question the United States faces is not whether Europe will develop robust AI capabilities. It is whether the United States will contribute to that development to strengthen itself and the Western bloc or stand aside while Europe advances more slowly in relative isolation—a trajectory that serves U.S. adversaries. The answer, however, depends fundamentally on how AI technology itself evolves.

Technological dominance without trust creates political pressure for independence. The EU's investments in governance, frontier models, and sectoral applications signal not merely a drive for technological sovereignty but also responses to perceived technological subordination. The same imbalances that provide the United States with leverage today are simultaneously driving allies toward options that will erode that advantage tomorrow. The relevant unit of analysis is no longer the nation-state but the technological ecosystem, and the United States must decide whether to cultivate a robust global AI alliance or defend national advantages until circumstances determine whether those advantages remain strategically valuable or become hollow.

The asymmetric nature of current U.S. and EU member state AI capacity creates a window of opportunity to structure cooperation on favorable terms. EU member states' reliance on U.S. technological infrastructure provides leverage that can be operationalized to benefit both parties while ensuring continued democratic leadership in AI. The timeline for that window's closure is uncertain. EU structural constraints are deep and slow-moving and Chinese alternatives remain incomplete, but the EU's direction toward sovereign AI ecosystems is accelerating. As EU member state capacity matures and alternative providers continue to emerge, the terms available to the United States for structuring engagements will narrow incrementally. Most importantly, the stakes extend beyond bilateral cooperation to the fundamental architecture of global AI development. In an era of competition with authoritarian powers, democratic nations may not be able to afford fragmented AI ecosystems that dissipate collective advantages.

The United States stands at an inflection point, where decisions about transatlantic AI partnership will shape the AI development landscape for decades. Success requires abandoning the comfortable fiction that transatlantic AI cooperation is a diplomatic courtesy extended to traditional allies. Success equally requires avoiding the assumption that partnership is uniformly beneficial regardless of technological developments. Both errors are, at root, failures of political ambition—the first is too modest, the second is insufficiently disciplined—and Gray's 2019 reminder that the choice of ends is the most consequential decision in strategic planning applies with full force today to the transatlantic AI relationship. The framework presented in this report is designed to navigate between these errors: pursue foundational efforts whose returns are scenario-invariant now and build institutional readiness for the scenario-contingent efforts whose value depends on which trajectory materializes while monitoring the signposts that indicate when to scale engagement or hold it at its current level.

Such is my argument for *conditional preparedness*—the deliberate construction of an architecture that allows the alliance to capture value from cooperation along whichever trajectory arrives, without overcommitting along those trajectories that do not. The price of that preparedness is manageable; the cost of discovering belatedly that the architecture was never built is not.

Abbreviations

AGI	artificial general intelligence
AI	artificial intelligence
CAISI	Center for AI Standards and Innovation
EIB	European Investment Bank
EU	European Union
EuroHPC JU	European High-Performance Computing Joint Undertaking
GDPR	General Data Protection Regulation
GPU	graphic processing unit
High-NA EUV	high numerical aperture extreme ultraviolet
IPCEI	Important Projects of Common European Interest
IT	information technology
NIST	National Institute of Standards and Technology
OECD	Organisation for Economic Co-operation and Development
PET	privacy-enhancing technology
R&D	research and development
SMEs	small- to medium-sized enterprises
STIC	Strategic Technology Investors Council
TSMC	Taiwan Semiconductor Manufacturing Company
TTC	Trade and Technology Council
UK	United Kingdom

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